
Why Linear Interpretability Methods Work: Invariant Subspaces from Architectural Constraints

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Abstract

Transformer representations admit reliable linear access through probes, sparse autoencoders, and activation steering, despite the depth and nonlinearity of the underlying networks. We give an architectural-statistical account of why. Architecturally, linear unembedding makes every logit margin between two output tokens a projection of the residual stream onto a fixed contrast direction, and the cross-entropy gradient pushes representations along weighted combinations of the same directions; under nonlinear readouts, this necessity disappears. Statistically, under sparse low-coherence superposition, projection onto fixed feature directions recovers feature magnitudes up to an explicit interference bound. A direct consequence is the *Self-Reference Property*: class prompts are themselves contexts that express a feature, so their activations provide approximate references for that feature direction without supervision. We validate the framework across eight classification tasks, four model families, and three architectures (decoder-only LLMs, BERT, ViT/LLaVA), with extensions to cross-modal, compositional tasks and attribution patching. Linear interpretability emerges as a predictable consequence of how transformers compute—fixed read and write directions modulated by context-dependent magnitudes—rather than an unexplained empirical regularity.

1 Introduction

Linear structure has become a central organizing principle in modern interpretability for transformers [Vaswani et al., 2017]. Linear probes recover semantic attributes from hidden states [Alain and Bengio, 2016, Belinkov, 2022], sparse autoencoders (SAE) identify interpretable feature directions [Bricken et al., 2023, Cunningham et al., 2023], and single-vector activation steering reliably modifies model behavior [Turner et al., 2023, Zou et al., 2023]. Across a wide range of methods and objectives, simple linear operations repeatedly succeed at isolating meaningful internal structures.

Yet this success raises a fundamental theoretical puzzle. Transformers are deep, highly nonlinear systems trained with simple objectives but exhibiting complex emergent behavior. Why, then, should their internal representations admit such simple and reliable linear access to semantic information?

We propose that transformer representations are organized as a system of fixed feature directions whose magnitudes vary with context. Each semantic feature corresponds to a stable direction in residual space; the activation for any context decomposes as a sparse sum over these directions; and across all contexts expressing a given feature, the direction stays the same—only the scalar magnitude varies. Linear interpretability methods recover meaningful structure because they read the same directional quantities the architecture itself reads. This analysis is not a claim about *why*

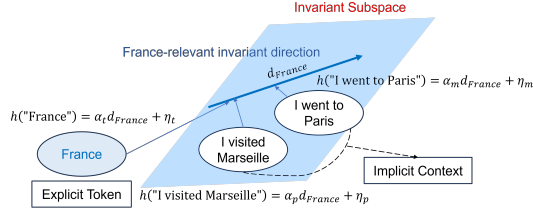


Figure 1: **Context-invariant directional representation.** The explicit token “France” provides a reference vector for this direction (*self-reference*), while contextual mentions such as “I went to Paris” and “I visited Marseille” share the same invariant direction.

features emerge—emergence depends on optimization dynamics, training data, and inductive biases that existing work studies—but about the geometric *form* reused features take whenever they are exposed through the model’s fixed linear interfaces.

Architecturally, linear unembedding makes every logit margin a projection of the residual stream onto a fixed contrast direction (Proposition 3.1); under nonlinear readouts, this necessity disappears (Proposition 3.2), so linearity is a property of the readout interface, not of cross-entropy alone. Statistically, sparse low-coherence superposition makes feature magnitudes recoverable by projection up to an explicit interference bound (Theorem 3.3). A direct consequence is the *Self-Reference Property*: a class prompt is itself a context where the feature is active, so its activation provides an approximate reference for that feature’s direction in a single forward pass, with no labeled instance data or learned probes (Corollary A.29). The activation produced by “France” is approximately aligned with the activation produced by “the country of the Eiffel Tower”; either can serve as a reference for locating the feature in other representations.

Our contributions are: (1) an architectural-statistical account of fixed-direction feature structure, with the readout’s role isolated by the contrast between linear and nonlinear unembeddings; (2) the Self-Reference Property as a zero-shot, label-free probe; (3) convergent empirical evidence across eight semantic classification tasks, four model families (LLaMA3-8B, Mistral-7B, GPT2-Small, LLaMA3.2-3B), and three architectures (decoder-only LLMs, BERT, ViT/LLaVA), including SAE-recovered features that align with class-token directions.

2 Related Work

Linear representation and probes. Semantic attributes are often captured by low-dimensional linear structure, dating back to regularities in word embeddings [Mikolov et al., 2013, Pennington et al., 2014], with modern operationalizations as *measurement*—linear probes that extract a property [Alain and Bengio, 2016, Hewitt and Manning, 2019, Tenney et al., 2019, Kim et al., 2018, nostalgebraist, 2020, Belrose et al., 2023, Kantamneni et al., 2025]—and *intervention*—steering along a direction [Ravfogel et al., 2020, Turner et al., 2023, Li et al., 2023, Rimsky et al., 2024, Zou et al., 2023]. Park et al. [2023] formalize what “linear representation” means and how measurement and intervention relate geometrically. Mechanistic interpretability builds on this to identify internal features, circuits, and causal mechanisms [Elhage et al., 2021, Nanda et al., 2023, Meng et al., 2022, Wang et al., 2022]. These results highlight the effectiveness of linear readouts but remain largely descriptive; our theory addresses why such decodability should arise in deep nonlinear models.

Relation to optimization-based accounts. Jiang et al. [2024] show that next-token prediction with softmax cross-entropy and the implicit bias of gradient descent can promote linear concept representations. Their setting assumes a linear unembedding; we make this dependence load-bearing. Proposition 3.2 shows that without a linear readout, the same task admits solutions whose target features are not linearly decodable, and the grokking ablation (Section 5.1) confirms the distinction empirically. Optimization supplies the pressure to learn task-relevant features; the readout architecture determines the coordinates through which that pressure operates.

Related invariant-subspace perspectives. Schiffman [2026] extract *algorithmic cores*—compact subspaces necessary and sufficient for task performance—that recur across independently trained transformers and across scales. Ahmad et al. [2025] decompose attention heads and MLPs into

orthogonal singular directions and find that canonical functional heads encode multiple overlapping subfunctions aligned with distinct singular directions. Where these works empirically identify low-dimensional structure, we provide an architectural and statistical account of *why* such structure must take a fixed-direction form whenever it is consumed by linear interfaces.

3 Theoretical Framework

We propose that transformer representations are organized as a system of fixed feature directions modulated by context-dependent magnitudes: each feature f corresponds to a stable direction $\mathbf{d}_f$ in residual space, and across all contexts expressing f , the direction is the same—only the magnitude $\alpha_f(c)$ varies. The transformer’s read and write operations are linear maps acting on this representation, so linear interpretability succeeds because it reads the same scalar magnitudes the architecture itself reads. This is a claim about the geometric *form* reused features take when exposed through the model’s fixed linear interfaces, not about which features emerge during training.

Setup. We consider standard decoder-only transformers with an additive residual stream, fixed linear interfaces (attention W_Q, W_K, W_V, W_O , and unembedding W_U), and a linear output layer $\ell(c) = W_U \mathbf{h}^{(L)}(c)$. We write $\mathbf{u}_t = (W_U)_t$ for the unembedding row of token t , and $\mathbf{h}(c) := \mathbf{h}^{(L)}(c)$ when no ambiguity arises. Throughout, we adopt the sparse-superposition representation model from prior work [Elhage et al., 2022]: residual activations decompose over a unit-norm dictionary $\mathcal{D} = \{\mathbf{d}_g\}_{g \in \mathcal{F}}$ with coherence $\mu = \max_{g \neq g'} |\langle \mathbf{d}_g, \mathbf{d}_{g'} \rangle|$, sparse support $|S(c)| \leq k$, and bounded residual $\|\xi(c)\| \leq \varepsilon$. We further assume that unembedding rows admit an analogous bundle decomposition $\mathbf{u}_t = \sum_{f \in B(t)} b_{t,f} \mathbf{d}_f + \mathbf{r}_t$ with $\|\mathbf{r}_t\| \leq \rho_t$, and that the architectural readout direction $\mathbf{w}_f = L_f^\top \phi_f$ aligns with the dictionary direction up to a misalignment parameter γ , $\langle \mathbf{w}_f, \mathbf{d}_f \rangle / \|\mathbf{w}_f\| \geq 1 - \gamma$. Full statements, motivation, and all proofs appear in Appendix A.

3.1 Architectural content: fixed contrast coordinates

The first content of the framework is that linear unembedding makes both inference and training operate through the same set of fixed coordinates.

Proposition 3.1 (Linear Unembedding Imposes Fixed Contrast Coordinates). *With logits $\ell(c) = W_U \mathbf{h}(c)$, the logit margin between any two output tokens is $\ell_y(c) - \ell_j(c) = (\mathbf{u}_y - \mathbf{u}_j)^\top \mathbf{h}(c)$, the projection of the residual stream onto the fixed contrast direction $\mathbf{v}_{y,j} := \mathbf{u}_y - \mathbf{u}_j$. For cross-entropy training with target token y and softmax probabilities $\mathbf{p}(c) = \text{softmax}(\ell(c))$, the gradient at the final residual stream is*

$$\nabla_{\mathbf{h}} \mathcal{L}(c) = W_U^\top (\mathbf{p}(c) - \mathbf{e}_y) = \sum_{j \neq y} p_j(c) (\mathbf{u}_j - \mathbf{u}_y),$$

a probability-weighted sum of the same fixed contrast directions that govern inference.

The architectural premise is the linearity of the unembedding—a fact about standard transformer design, not an assumption about features. The conclusion is that fixed contrast directions $\mathbf{u}_y - \mathbf{u}_j$ govern both inference (which token the model picks) and training (how the residual stream is updated). The proposition does not claim semantic features must emerge; it states that *if* a feature affects the model’s choice between two tokens, it does so through the corresponding fixed contrast coordinate.

The following counterfactual isolates the readout’s role: under nonlinear readouts, fixed-coordinate structure is no longer required.

Proposition 3.2 (Nonlinear Readouts Remove Fixed-Coordinate Necessity). *Let logits be produced by a nonlinear readout $\ell = G(\mathbf{h})$, and define the margin between output tokens y and j as $m_{y,j}(\mathbf{h}) = G_y(\mathbf{h}) - G_j(\mathbf{h})$. Then*

$$\nabla_{\mathbf{h}} m_{y,j}(\mathbf{h}) = \nabla_{\mathbf{h}} G_y(\mathbf{h}) - \nabla_{\mathbf{h}} G_j(\mathbf{h}),$$

which depends on $\mathbf{h}$ in general. Unless this gradient is constant over the relevant domain of $\mathbf{h}$, no fixed direction $\mathbf{v}_{y,j}$ and scalar b exist with $m_{y,j}(\mathbf{h}) = \mathbf{v}_{y,j}^\top \mathbf{h} + b$. Nonlinear readouts therefore remove the architectural necessity that output-relevant evidence be exposed through a single fixed contrast coordinate; the same task can in principle be solved by representations whose target features are not linearly decodable.

Proof. A function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is affine ($f(\mathbf{h}) = \mathbf{v}^\top \mathbf{h} + b$ for fixed $\mathbf{v}, b$) if and only if $\nabla_{\mathbf{h}} f$ is constant. The margin gradient $\nabla_{\mathbf{h}} m_{y,j} = \nabla_{\mathbf{h}} G_y - \nabla_{\mathbf{h}} G_j$ is constant in $\mathbf{h}$ if and only if both G_y and G_j are affine up to a common nonlinear part that cancels in the difference. For generic nonlinear G (e.g., a deep MLP), this fails, and $m_{y,j}$ is not affine. $\square$

Together, Propositions 3.1 and 3.2 isolate the architectural contribution: linear readouts make fixed contrast coordinates necessary; nonlinear readouts do not. This makes visible a conditioning often left implicit: prior work showing that linear concept representations emerge under cross-entropy training [Jiang et al., 2024] assumes a linear unembedding as a fixed background condition. The linearity of features in standard transformers is not a consequence of cross-entropy alone, but of cross-entropy applied through linear or selection-only readout interfaces—a distinction the grokking ablation (Section 5.1) confirms empirically.

The same interface-local analysis applies to attention: W_Q, W_K, W_V read the residual stream along fixed directions, while the input-dependent softmax determines *how much* content is routed across positions but not *which* directions are read. Features consumed by an attention head are therefore exposed at that head’s input through the same fixed-direction structure that governs the unembedding, locally to that head rather than globally. The 2-layer MLP gradient analysis is treated in Appendix A (Proposition A.1).

3.2 Statistical content: projection error and its decomposition

Architectural pressure on directions is necessary but not sufficient for projection-based interpretability to work: the recovered scalar must actually approximate the underlying feature magnitude. We characterize the error of projection-based recovery exactly, identifying two interference terms that govern when the bound is tight.

Theorem 3.3 (Projection Error Decomposition). *Let $\mathbf{h}(c) = \alpha_f(c) \mathbf{d}_f + \sum_{g \in S(c), g \neq f} \alpha_g(c) \mathbf{d}_g + \boldsymbol{\xi}(c)$ with $\|\mathbf{d}_f\| = 1$. Then projection onto $\mathbf{d}_f$ decomposes exactly as*

$$\langle \mathbf{h}(c), \mathbf{d}_f \rangle = \alpha_f(c) + I_f(c) + R_f(c),$$

where

$$I_f(c) := \sum_{g \in S(c), g \neq f} \alpha_g(c) \langle \mathbf{d}_g, \mathbf{d}_f \rangle$$

is the cross-feature interference and $R_f(c) := \langle \boldsymbol{\xi}(c), \mathbf{d}_f \rangle$ is the residual contamination. Therefore

$$|\langle \mathbf{h}(c), \mathbf{d}_f \rangle - \alpha_f(c)| \leq |I_f(c)| + |R_f(c)|.$$

If additionally $|\langle \mathbf{d}_g, \mathbf{d}_f \rangle| \leq \mu$ for $g \neq f$, $|S(c)| \leq k$, $|\alpha_g(c)| \leq A$, and $\|\boldsymbol{\xi}(c)\| \leq \varepsilon$, then the worst-case bound

$$|\langle \mathbf{h}(c), \mathbf{d}_f \rangle - \alpha_f(c)| \leq \mu(k-1)A + \varepsilon$$

follows. Projection onto the architectural readout direction $\mathbf{w}_f$ admits the same decomposition with an additional $\sqrt{2\gamma} \|\mathbf{h}(c)\|$ term from readout–dictionary misalignment.

The first bound is the main content: projection succeeds exactly when cross-feature interference $I_f(c)$ and residual contamination $R_f(c)$ are small. Sparsity and low-coherence are *one sufficient way* to make these terms small, yielding the worst-case $\mu(k-1)A + \varepsilon$ expression standard in compressed-sensing analyses; they are not the only way. Crucially, $I_f(c)$ and $R_f(c)$ are individually empirically measurable: the SAE overlap analysis in Section 4 estimates the cross-feature term by measuring the support overlap between class-prompt-active and instance-active SAE latents, and the wrong-class projection mass tracks how much $\langle \mathbf{h}(c), \mathbf{d}_j \rangle$ deviates from zero for $j \neq y$ on instances of class y (a direct estimate of $|I_j(c)|$ for non-active directions). The theorem therefore provides a diagnostic decomposition rather than a black-box assumption: when projection-based recovery succeeds or fails, the framework predicts which interference term is responsible.

Combining the architectural and statistical content gives a concrete account of next-token prediction. Expanding $\mathbf{u}_t^\top \mathbf{h}(c)$ over both decompositions, the matching part isolates the bundle features active in context:

$$\ell_t(c) = \mathbf{u}_t^\top \mathbf{h}(c) = \sum_{f \in B(t) \cap S(c)} b_{t,f} \alpha_f(c) + \Delta_t(c),$$

with $\Delta_t(c)$ bounded by the same interference terms as Theorem 3.3 (Proposition A.1, Appendix A). Next-token prediction reduces to weighted feature-bundle matching against the active features in the residual stream.

Corollary 3.4 (Additive Feature Contributions and Intervention Predictions). *Under the bundle-matching decomposition above, the contribution of feature f to logit $\ell_t(c)$ is $b_{t,f} \alpha_f(c)$; the contribution of a group $G \subseteq S(c)$ is $\sum_{f \in G \cap B(t)} b_{t,f} \alpha_f(c)$; and an intervention setting $\alpha_f(c) \rightarrow \alpha_f^*$ produces a logit change of*

$$\Delta \ell_t = b_{t,f} (\alpha_f^* - \alpha_f(c)) + \Delta_t^{\text{int}}(c),$$

where $\Delta_t^{\text{int}}(c)$ is bounded by the same interference terms of Theorem 3.3.

The corollary makes intervention predictions quantitative: suppressing f along $\mathbf{d}_f$ should change the logit margin by $b_{t,f} \alpha_f(c)$, suppressing a group should produce additive effects, and across instances of class y larger $\alpha_y(c)$ should produce proportionally larger logit margins for y . We do not run intervention experiments here; the framework specifies what they should produce, and Section 5 measures the underlying $\alpha_f(c)$ that interventions would target.

3.3 The Self-Reference Property

A direct consequence is that class prompts can serve as approximate references for feature directions without supervision. A class prompt t_f —a token or expression canonically expressing feature f —is itself a context where f is active. Its activation, which we write $\mathbf{h}_{t_f}$, therefore decomposes as $\mathbf{h}_{t_f} = \beta_f \mathbf{d}_f + \boldsymbol{\nu}_f$ with $\beta_f > 0$ (the class-prompt magnitude) and $\|\boldsymbol{\nu}_f\| \leq \delta_f$ (residual contamination from other components active in the prompt). Up to the contamination ratio δ_f/β_f , the class-prompt activation is aligned with $\mathbf{d}_f$ and can be used in its place to detect f in any other context.

Theorem 3.5 (Self-Reference Classification Margin). *Let $\{\mathbf{d}_1, \dots, \mathbf{d}_K\}$ be unit class-feature directions with cross-class coherence $|\langle \mathbf{d}_i, \mathbf{d}_j \rangle| \leq \mu_C$ for $i \neq j$. Suppose the class prompt activation for class k decomposes as $\mathbf{h}_{t_k} = \beta_k \mathbf{d}_k + \boldsymbol{\nu}_k$ with $\beta_k > 0$ and $\|\boldsymbol{\nu}_k\| \leq \delta_k$, and an instance c of class y decomposes as $\mathbf{h}(c) = \alpha_y(c) \mathbf{d}_y + \boldsymbol{\eta}(c)$ with $\alpha_y(c) > \tau_y$ and $|\langle \mathbf{d}_k, \boldsymbol{\eta}(c) \rangle| \leq \tau_k$ for every k . Then the self-reference classifier $\hat{y}(c) = \arg \max_{k \in [K]} \mathbf{h}_{t_k}^\top \mathbf{h}(c)$ is correct if, for every $j \neq y$,*

$$\beta_y (\alpha_y(c) - \tau_y) - \delta_y \|\mathbf{h}(c)\| > \beta_j (\mu_C \alpha_y(c) + \tau_j) + \delta_j \|\mathbf{h}(c)\|.$$

Each term in the margin condition maps to a measurable quantity: $\alpha_y(c)$ is the class signal strength, μ_C is cross-class direction coherence, τ_j is instance interference along class j ’s direction, and δ_j is class-prompt contamination. Failure modes correspond to violations of specific terms (low signal, overlapping classes, contaminated prompts), each of which is empirically diagnosable. This is the central theorem behind the experimental section: the class-prompt classifier is exactly the empirical instantiation of $\mathbf{h}_{t_k}^\top \mathbf{h}(c)$.

The mechanism behind Self-Reference. The dictionary that organizes residual activations also organizes attention’s OV writes and the unembedding’s bundle scoring. When OV approximately preserves dictionary directions, attention aggregates per-feature evidence from prompt tokens into the prediction position, and the unembedding scores candidates by bundle matching against this aggregate (App. A). Surface forms expressing the same feature—“France,” “Paris,” “Eiffel Tower”—thus route through the same output-facing coordinate.

3.4 Extension to MLP blocks and LayerNorms

The framework extends to MLP-written features under a *selection-only feature decomposition*: outputs factor as $\sum_i g_i(\mathbf{h}) \mathbf{w}_i^{\text{out}}$ with fixed write directions $\mathbf{w}_i^{\text{out}}$ and per-unit gates g_i depending only on each unit’s own read direction. Standard 2-layer MLPs (LLaMA, GPT, Mistral) satisfy this structurally; deeper MLPs break it through compositional gating (Proposition A.27, App. A). LayerNorm preserves directional invariance: feature directions map to transformed feature directions with context entering only through a scalar rescaling of $\alpha_f(c)$ (Proposition A.22). Attention OV, the unembedding, 2-layer MLPs, and LayerNorm thus all write or transform through fixed directions modulated by context-dependent scalar gates.

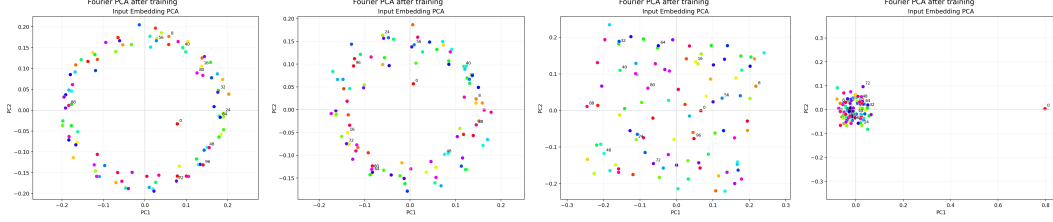


Figure 2: Input embedding PCA after training on modular division ($p=97$), for readout heads with 1, 2, 3, and 4 layers (left to right). Canonical positions for the Fourier solution are labeled. With a linear readout and a 2-layer MLP, embeddings form the predicted circular structure exactly. Adding more nonlinearities progressively removes the geometric constraint.

4 Methods for Extracting Directions

The Self-Reference Property (Corollary A.29) implies that a classifier built on class-prompt directions should generalize to instance prompts that express the same feature, since both routes land on the same invariant direction. We compare three approaches that leverage this insight, all of which classify by projecting attention-head activations onto normalized class directions; they differ only in how directions are extracted. We evaluate each attention head independently.

Zero-Shot Probe. Use class-token hidden states as class directions, mean-centered to remove components shared across all class prompts (e.g., “being a country”):

$$\hat{\mathbf{d}}_k = \frac{\mathbf{h}_{t_k} - \bar{\mathbf{h}}}{\|\mathbf{h}_{t_k} - \bar{\mathbf{h}}\|}, \quad \bar{\mathbf{h}} = \frac{1}{K} \sum_{j=1}^K \mathbf{h}_{t_j}. \quad (1)$$

No training data required.

Unsupervised Probe. A lightweight linear transformation W trained on *class prompts* $\mathbf{h}_{t_k}$ only (no instance labels) via a contrastive objective. With $\mathbf{c}_k := W\mathbf{h}_{t_k}$ and $S = \mathbf{C}\mathbf{C}^\top/\tau$,

$$\mathcal{L} = - \sum_{k=1}^K \log \frac{\exp(S_{kk})}{\sum_{j=1}^K \exp(S_{kj})}. \quad (2)$$

The Self-Reference Property predicts W should generalize: if tokens and instances share invariant directions, orthogonalizing tokens automatically orthogonalizes instances.

Sparse Autoencoder Probe. We train sparse autoencoders (SAEs) on attention-head outputs [Cunningham et al., 2023, Bricken et al., 2023, Kissane et al., 2024], producing sparse latents $\mathbf{z} = \text{TopK}(W_{\text{enc}}\mathbf{h} + \mathbf{b}_{\text{enc}})$. SAEs are trained only on instance activations (no class tokens); we then check whether class-token latents and instance latents activate the same SAE dimensions.

Unified Classification. Letting $\phi(\cdot)$ denote the chosen transformation (mean-centering, W , or SAE encoder), classification uses the Identity-Projection operator with normalized class vectors:

$$\hat{k} = \arg \max_{k \in [K]} \phi(\mathbf{h}(c))^\top \hat{\phi}_k, \quad \hat{\phi}_k = \frac{\phi(\mathbf{h}_{t_k})}{\|\phi(\mathbf{h}_{t_k})\|}. \quad (3)$$

The query is not normalized, so projection magnitude reflects feature strength.

5 Experiments

5.1 Architectural validation: linear vs. nonlinear readouts

Proposition 3.2 predicts that fixed-direction structure is architecturally necessary under linear readouts and merely permissible under nonlinear ones. Prior work on grokking [Power et al., 2022] shows that

Table 1: Classification accuracy (%) by method and model. Unsupervised: top-5 head mean; SAE, Zero-Shot, and Text Output: single-run top-5 head mean. The bottom block probes MLP activations at the last-token position; all other rows probe attention-head outputs. SBERT [Reimers and Gurevych, 2019] (all-mpnet-base-v2) cosine is included as a label-free external baseline.

Model	Method	Top-5 head mean Accuracy (%)							
		Animals	Countries	C. Chars	Authors	Langs	Emotions	Fruits	Companies
SBERT		70.0	78.46	27.67	24.37	85.33	57.43	33.5	58.0
LLaMA3-8B	SAE	80.15	70.97	52.22	63.13	87.41	36.91	57.10	73.04
	Unsupervised	89.20	77.10	56.43	69.75	85.70	51.58	62.36	78.06
	Zero-Shot	76.85	76.69	51.20	57.27	84.81	49.44	55.87	74.71
	Text Output	99.67	79.48	42.00	68.75	94.00	68.40	49.00	88.50
Llama3.2-3B	SAE	71.96	76.06	43.82	41.68	87.21	34.00	48.20	67.10
	Unsupervised	84.58	73.67	43.02	52.25	87.54	48.22	57.30	74.00
	Zero-Shot	65.87	72.78	40.92	47.73	87.82	42.40	50.18	70.93
	Text Output	97.67	46.15	33.33	39.37	62.50	62.20	51.00	88.50
Mistral	SAE	73.61	60.47	46.14	39.49	91.43	39.44	52.00	74.59
	Unsupervised	84.14	64.68	46.10	49.51	90.74	46.82	56.67	81.34
	Zero-Shot	65.57	64.17	41.64	43.86	92.61	45.25	49.05	78.33
	Text Output	92.33	33.85	33.33	40.00	98.67	75.20	84.50	89.00
GPT2-Small	SAE	21.74	29.48	19.14	13.72	62.79	16.87	29.64	40.68
	Unsupervised	24.14	37.83	21.16	19.19	72.54	18.81	34.63	51.70
	Zero-Shot	21.27	32.67	19.44	17.80	64.94	16.74	30.40	50.32
	Text Output	35.33	25.64	17.00	12.50	15.67	49.80	37.50	29.50
LLaMA3-8B (MLP)	Unsupervised	66.19	52.97	43.37	66.96	36.99	21.38	44.77	64.23
	Zero-Shot	58.86	52.23	40.58	56.74	43.06	20.19	42.73	57.24

transformers learning modular arithmetic develop Fourier representations—embeddings arranged in a circle where addition becomes rotation. We test whether this structure is forced by the task or by the readout: if linear, Fourier structure should be architecturally required; if nonlinear, it should be one solution among many.

We train a 2-layer transformer on modular division ($p=97$) and vary the readout head’s nonlinearity from 1 to 4 layers, training a separate linear probe on the frozen hidden states across 10 seeds and 50000 steps per configuration. With a fully linear head, the linear probe matches the head exactly: both reach 1.00 ± 0.00 across all seeds, and the input embeddings form a clean Fourier circle (Fig. 2, leftmost). With 2 layers, the MLP head still averages 0.98 ± 0.02 and the linear probe keeps a high accuracy with 0.95 ± 0.08 and the model still finds the Fourier structure for it. With 3 layers, the head retains 0.97 ± 0.05 accuracy, while the linear probe collapses uniformly to 0.19 ± 0.10 , and the circular geometry is largely absent. With 4 layers the embedding structure degenerates – tokens collapse to a tight cluster near the origin, with no angular organization at all, the MLP learns to generalize correctly with an accuracy of 0.95 ± 0.3 , while the linear probe completely fails with 0.03 ± 0.01 accuracy. Fourier representations are therefore not inevitable under cross-entropy training; they form when and only when the readout requires them, exactly as Proposition 3.2 predicts.

5.2 Classification Results

We evaluate mainly across all the attention head outputs, on eight classification tasks spanning diverse semantic domains (Table 6): taxonomic (Animals), geographic (Countries), affective (Emotional Sentences [Ghazi et al., 2015]), stylistic (Literary Quotes, Cartoon Phrases), and linguistic (Languages [Artetxe et al., 2020]) and descriptive (Fruits, Companies).

For each task, we obtained a list of *implicit* sentences that express the class without mentioning it (e.g., “the country of the Eiffel Tower”). The full description of the datasets can be found in the Appendix F.1.

Table 1 reports per-task classification accuracy. We highlight three findings.

Self-Reference holds across most heads. Across four datasets in LLaMA3-8B, 86–98% of attention heads exhibit within-class alignment exceeding between-class alignment (App. E.1, Fig. 7), confirming the Self-Reference Property holds at population level. We report top-5 head accuracy in Table 1 because the framework predicts feature-specific readout directions in the heads that implement each feature, not uniform recoverability—most heads do positional, syntactic, or copying work

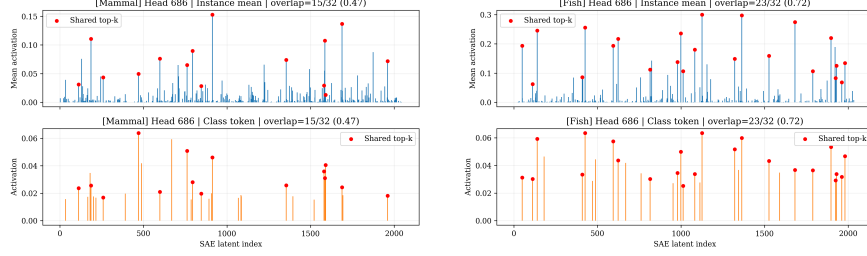


Figure 3: SAE shared-peak analysis. Top- k SAE dimensions of a class token compared with the top- k most frequent SAE dimensions across instances of that class. Red markers denote shared dimensions.

unrelated to any given semantic feature. Examples of the full per-head accuracy distribution appear in App. D.1, Fig. 6.

All three methods recover the predicted structure. Zero-shot, unsupervised, and SAE-based classification capture the same set of attention heads and distinguish between classes of similar semantic meaning, providing converging evidence across methods that differ only in how directions are extracted—from tokens directly, learned orthogonalization, or sparse decomposition. The unsupervised method consistently outperforms zero-shot, indicating that learned orthogonalization improves separability by disentangling shared category components (“being a country”) from class-specific ones. SAEs are the strongest result of the three: trained only on instance activations, never on class tokens, they still reach competitive accuracy—direct evidence that class-token directions coincide with the dictionary atoms organizing instance representations (Section 5.3).

Generalization across architectures and modalities. Consistent performance across LLaMA3-8B, Mistral-7B, GPT2-Small, and LLaMA3.2-3B suggests directional invariance is not an artifact of any particular family or scale. The bottom block of Table 1 probes MLP activations at the last-token position: attention consistently outperforms MLP on categorical tasks (Animals, Countries, Languages, Emotions), but the gap narrows on more compositional or stylistic tasks (Authors, Characters, Companies). MLP outputs being decodable above chance is consistent with the framework (Section 3).

5.3 SAE-overlap analysis: tokens activate the same features as instances

To test whether class tokens and their semantic instances activate the *same* SAE features, we trained one sparse autoencoder per attention head following Kissane et al. [2024], using only instance activations—class tokens were never seen during training. Despite this, class-token activations and instance activations consistently activate overlapping top- k SAE dimensions. Figure 3 illustrates the animal dataset: with $k=32$, mammals share 15/32 features, fish 23/32, reptiles 25/32, birds 22/32. The pattern holds across datasets and model sizes, and provides direct evidence on the cross-feature interference term $I_f(c)$ in Theorem 3.3: the same dictionary atoms that organize residual activations across instances also organize the activations elicited by class prompts.

5.4 Activation Patching

To validate Corollary 3.4, we use the Zero-Shot Probe approach extended to two contrastive prompts and two classes. Given a clean and corrupted prompt pair $(p_{\text{clean}}, p_{\text{corr}})$ with corresponding answers $(a_{\text{clean}}, a_{\text{corr}})$, we extract the class-direction

$$\Delta a = h(a_{\text{clean}}) - h(a_{\text{corr}}) \in \mathbb{R}^{d_{\text{model}}} \quad (4)$$

from the residual stream at the last layer’s output (SRP-BP) or from the component’s space (SRP). For SRP-BP, since Δa lives in $\mathbb{R}^{d_{\text{model}}}$ but each component’s activation lives in its own native space, we project Δa into each component’s native space through the architecture’s output matrix— $W_O^{\ell, h}$ for attention head h at layer ℓ , W_{out}^{ℓ} for the MLP at layer ℓ —and rank components by alignment with

Method	Faithfulness			Completeness		
	$k=1$	$k=3$	AUC	$k=1$	$k=3$	AUC
<i>IOI</i>						
AtP	-0.43	-0.10	0.970	-0.37	-0.08	0.974
AtP*	0.17	0.86	0.990	0.16	0.85	0.993
SRP	0.17	-0.10	0.965	0.16	-0.08	0.970
SRP-BP	0.86	1.09	0.995	0.85	1.10	0.999
<i>Capitals</i>						
AtP	0.31	0.97	0.983	0.31	0.96	0.986
AtP*	0.82	0.97	0.987	0.81	0.96	0.990
SRP	0.34	0.97	0.986	0.34	0.96	0.988
SRP-BP	0.82	0.97	0.990	0.81	0.96	0.990

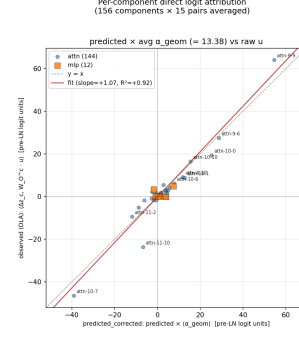


Figure 4: Projection-based attribution on GPT-2 small. **Left:** Faithfulness and completeness at $k = 1, 3$ and AUC across $k \in \{1, 2, 3, 8, 16, 31, 78, 156\}$ for IOI and Capitals. On Capitals, AtP/SRP rank attention heads first while AtP*/SRP-BP rank mlp-0 first. **Right:** Predicted contributions (Corollary 3.4), rescaled by the per-pair geometric factor α_{geom} (mean 14.59), versus observed DLA $\langle \Delta z_c, W_O^c u \rangle$ on IOI components.

the prompt-side activation difference at that component:

$$d^c = W^c \frac{\Delta a}{\|\Delta a\|} \in \mathbb{R}^d, \quad s^c = \langle z^c(p_{\text{clean}}) - z^c(p_{\text{corr}}), d^c \rangle, \quad (5)$$

where z^c denotes the component’s activation in its native space ($z_{\ell,h} \in \mathbb{R}^{d_{\text{head}}}$ for attention heads, $m_{\ell} \in \mathbb{R}^{d_{\text{mlp}}}$ for MLPs). Finally, we patch the top- k components from clean to corrupted and measure how much of the logit-difference metric is recovered, comparing against gradient-based attribution patching (AtP [Nanda, 2023]) and its GradDrop variant (AtP* [Kramár et al., 2024]) as baselines.

Figure 4 compares the four methods on GPT2-Small for IOI and Capitals. SRP-BP outperforms AtP, AtP*, and SRP on IOI [Wang et al., 2022], where AtP and SRP go negative at $k=1, 3$ because negative name-mover heads dominate gradient-based scores; on Capitals, SRP-BP performs comparably to AtP*. To test the quantitative prediction of Corollary 3.4, we compare the per-component magnitudes predicted by SRP-BP against the observed direct logit attribution (DLA, Elhage et al. 2021) on the IOI components: predicted and observed values trace a line with slope 1.07 and $R^2 = 0.92$ (Fig. 4, right), validating the linear-path contribution claim that downstream effects scale with $\alpha_f(c)$.

Additional generalization. App. E.3 and E.4 extend the probe beyond decoder-only LMs and categorical classification. The same self-reference probe works on BERT [Devlin et al., 2019], ViT [Dosovitskiy et al., 2020], and LLaVA [Liu et al., 2023] (BERT Languages: 93.0%; ViT/LLaVA ImageNet-10 [Russakovsky et al., 2015]: 79.5%/78.6%), and remains above random on multi-hop, ICL, arithmetic, SVAMP [Patel et al., 2021], and refCOCO [Yu et al., 2016] tasks. These results support the view that fixed-direction exposure is tied to linear interfaces rather than to autoregressive language modeling alone.

6 Conclusion

We presented an architectural-statistical account of why linear interpretability works in transformers: linear readouts impose fixed contrast coordinates, nonlinear readouts remove that necessity, and sparse-superposition interference controls when projection recovers feature magnitudes. The Self-Reference Property turns this geometry into a label-free probe, using class prompts as feature references. Results across classification, SAE overlap, cross-architecture probes, and attribution patching support the view that reused output-relevant features are exposed through linearly readable coordinates.

Limitations. The framework covers features communicated through linear or selection-only interfaces, not features used only through input-dependent routing or deeper nonlinear paths. It also does not decide when invariant subspaces are one-dimensional, and its recoverability bounds are in-distribution rather than guarantees under compositional shift [Barin Pacela et al., 2026].

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A Detailed Theoretical Framework

This appendix provides the full theoretical framework underlying the compact main-body Section 3. We restate definitions and assumptions in detail, present supplementary corollaries and remarks (including the Identity-Projection operator and capacity-driven feature sharing), and discuss the bridges between the architectural readout direction $\mathbf{w}_f$ and the dictionary direction $\mathbf{d}_f$. All proofs of main-body results are given in Appendix B below.

A.1 Statement Restatements

For ease of reference, we restate here the main-body theorems and propositions whose detailed discussion appears below or whose proofs appear in Appendix B. The labels are duplicated on these restated versions so that appendix discussion and proofs can reference them naturally.

Lemma (Linear-Path Communication Lemma; restated). For a fixed linear interface $L : \mathbb{R}^d \rightarrow \mathbb{R}^{k_L}$ and fixed downstream readout $\phi \in \mathbb{R}^{k_L}$, defining the residual-stream direction $\mathbf{w} := L^\top \phi$,

$$\phi^\top L \mathbf{h}(c) = \mathbf{w}^\top \mathbf{h}(c) \quad \forall c.$$

Proposition (Nonlinear Readouts Relax Output-Facing Directional Pressure; restated). Let logits be produced by a (possibly nonlinear) readout $\ell(c) = G(\mathbf{h}^{(L)}(c))$. Then

$$\nabla_{\mathbf{h}^{(L)}} \mathcal{L}(c) = J_G(\mathbf{h}^{(L)}(c))^\top (\mathbf{p}(c) - \mathbf{e}_y),$$

where J_G is the Jacobian of the readout. The rows of J_G depend on c for general nonlinear G . For a 2-layer MLP $G(\mathbf{h}) = W_{\text{out}} \sigma(W_{\text{in}} \mathbf{h} + \mathbf{b})$, the gradient simplifies to a context-gated sum over the fixed columns of $W_{\text{in}}^\top$:

$$\nabla_{\mathbf{h}^{(L)}} \mathcal{L} = W_{\text{in}}^\top \text{diag}(\sigma'(\cdot)) W_{\text{out}}^\top (\mathbf{p}(c) - \mathbf{e}_y).$$

For deeper readouts, the Jacobian becomes a chain product with cross-coupled gates; see Remark A.9.

Proposition (Next-Token Prediction as Feature-Bundle Matching; restated). Under Assumptions A.12 and A.17, the logit assigned to token t in context c is

$$\ell_t(c) = \mathbf{u}_t^\top \mathbf{h}(c) = \sum_{f \in B(t) \cap S(c)} b_{t,f} \alpha_f(c) + \Delta_t(c),$$

where the interference term satisfies

$$|\Delta_t(c)| \leq \mu \sum_{f \in B(t)} |b_{t,f}| \sum_{\substack{g \in S(c) \\ g \neq f}} |\alpha_g(c)| + \|\mathbf{r}_t\| \|\mathbf{h}(c)\| + \|\mathbf{u}_t\| \|\boldsymbol{\xi}(c)\|.$$

Example 1 (Solvable without a linear feature direction). Let representations be $\mathbf{h} = (x_1, x_2) \in \{-1, 1\}^2$ and the target be $y = x_1 x_2$. The two classes are not linearly separable in $\mathbf{h}$, so no fixed direction $\mathbf{v}$ and scalar b recover y via projection $\mathbf{v}^\top \mathbf{h} + b$. A 2-layer nonlinear readout, however, can compute y exactly. Cross-entropy training of such a readout can therefore be minimized without producing a linear feature for y in the input representation $\mathbf{h}$. This is the simplest concrete instance of the non-necessity content of Proposition 3.2: a learnable mapping from inputs to a correct output that does not factor through any fixed contrast coordinate in the upstream representation.

A.2 Preliminaries and Assumptions

We consider decoder-only transformers with the following properties:

Assumption A.1 (Architectural Requirements). Let $\mathcal{M}$ be a transformer with L layers and hidden dimension d . We assume:

1. **Additive Residual Stream.** The residual stream updates additively: $\mathbf{r}^{(\ell+1)} = \mathbf{r}^{(\ell)} + \Delta^{(\ell)}$, where $\Delta^{(\ell)}$ is the output of layer ℓ .
2. **Linear Communication Interfaces.** The OV circuit in attention ($W_O W_V$), the QKV projections (W_Q, W_K, W_V), and the unembedding layer (W_U) act as linear maps on the residual stream.

3. **Shared Parameters.** The same parameters are applied across all token positions; optimization occurs over parameters shared across contexts.
4. **Linear Output Layer.** Token logits are computed via linear projection: $\text{logits} = W_U \mathbf{r}^{(L)}$.

These assumptions hold for standard transformer architectures including GPT-2, LLaMA, and similar models.

We denote by $\mathcal{L}$ the collection of read-side linear interfaces in the model:

$$\mathcal{L} = \{W_U\} \cup \{W_Q^{(\ell,h)}, W_K^{(\ell,h)}, W_V^{(\ell,h)}\}_{\ell,h} \quad (6)$$

indexed over layers ℓ and attention heads h . Each $L \in \mathcal{L}$ is a linear map $L : \mathbb{R}^d \rightarrow \mathbb{R}^{k_L}$ that is fixed at inference. The set $\mathcal{L}$ formalizes what we mean by “communication interfaces”: every linear map that reads from the residual stream and is shared across positions and contexts. We treat these as fixed maps that determine the *directions* along which content is read; the input-dependent dynamics of attention routing (the softmax of $Q^\top K / \sqrt{d}$) determine *how much* content is routed but do not modify these directions, and lie outside the scope of the feature-direction analysis.

A.3 Formal Definitions

Definition A.2 (Context). A *context* $c = (x_1, \dots, x_n; i)$ consists of an input token sequence and a position of interest i . We write $\mathbf{h}^{(\ell)}(c) \in \mathbb{R}^d$ for the hidden state at layer ℓ and position i in context c .

Definition A.3 (Semantic Feature). A *semantic feature* is a function $f : \mathcal{C} \rightarrow \mathcal{Y}$ mapping contexts to a value space $\mathcal{Y}$. For classification tasks, $\mathcal{Y} = \{y_1, \dots, y_K\}$ is finite.

Definition A.4 (Communicable Feature). A semantic feature f is *communicable through interface* $L \in \mathcal{L}$ along direction $\phi \in \mathbb{R}^{k_L}$ if it satisfies both:

1. **Multi-context:** There exist distinct contexts c_1, c_2 with $f(c_1) = f(c_2)$.
2. **Linear decodability:** The feature value is recoverable through L :

$$\phi^\top L \mathbf{h}(c) = g(f(c)) \quad \forall c \in \mathcal{C} \quad (7)$$

for some function $g : \mathcal{Y} \rightarrow \mathbb{R}$. Equivalently, $\mathbf{w}_f^\top \mathbf{h}(c) = g(f(c))$ where $\mathbf{w}_f := L^\top \phi \in \mathbb{R}^d$ is the *decoding direction* in the residual stream.

The multi-context requirement captures that meaningful features appear across different surface realizations (e.g., “France” and “the country of the Eiffel Tower” both express the France feature). Linear decodability through $L \in \mathcal{L}$ captures that the feature must be extractable by some downstream linear interface—most importantly the unembedding W_U and the attention QKV projections.

Definition A.5 (Invariant Subspace). A communicable feature f exhibits *contextual invariance* if there exists a subspace $\mathcal{S}_f \subseteq \mathbb{R}^d$, determined by f and model parameters alone, such that for all contexts c expressing f , the f -relevant information in $\mathbf{h}(c)$ lies entirely within $\mathcal{S}_f$.

Definition A.6 (Directional Invariance). A communicable feature f exhibits *directional invariance* if it exhibits contextual invariance with $\dim(\mathcal{S}_f) = 1$, i.e., $\mathcal{S}_f = \text{span}(\{\mathbf{d}_f\})$ for some direction $\mathbf{d}_f \in \mathbb{R}^d$.

A.4 Main Theoretical Results

The remainder of this section gives the architectural and statistical content of this analysis. The architectural content is twofold. First, the Linear-Path Communication Lemma (Lemma A.1) records the elementary identity that fixed downstream linear paths correspond to fixed residual-stream directions. Second, Proposition 3.1 traces this identity into output decisions and training: every logit margin between two output tokens equals the projection of the final residual stream onto a fixed contrast direction, and the cross-entropy gradient pushes representations along weighted combinations of the same contrast directions. Proposition A.1 sharpens the architectural specificity: a 2-layer MLP readout preserves fixed read directions through context-dependent scalar gates, while deeper compositions generically break this independent-gating factorization. The statistical content adds the conditions under which features are recoverable from data: sparse low-coherence feature

organization (Assumption A.12) and bounded readout–dictionary alignment (Assumption A.13) yield Theorem 3.3, which bounds the error of projection-based feature recovery. Combining these with a token-bundle decomposition of the unembedding (Assumption A.17) gives Proposition A.1 and Corollary A.18: next-token prediction reduces to weighted feature-bundle matching, and contributions to logit margins are read through fixed contrast directions. The Self-Reference Property follows: class prompts, being themselves contexts that express a feature, provide approximate references for its direction without supervision (Corollary A.29), with explicit margin conditions for the resulting classifier (Theorem 3.5).

The lemma’s proof is one line, and that is the right shape for a load-bearing identity: the architectural fact it records (fixed linear paths induce fixed directions) is what the rest of the framework leverages. The first such leveraging is at training time, through the cross-entropy gradient.

Interpretation: architecture and training share the same coordinates. The proposition indicates that the same fixed contrast directions $\mathbf{u}_y - \mathbf{u}_j$ govern both how the model decides between two tokens at inference and how training updates push the residual stream during learning. Cross-entropy rewards making the evidence for the correct token accessible through the same contrast coordinates the model uses to compare tokens. This connects optimization directly to the geometry of decisions, without requiring a separate assumption about what features are linearly decodable.

Remark A.7 (Scope: final residual stream, not arbitrary layers). Proposition 3.1 is stated for the gradient at the *final* residual stream $\mathbf{h}^{(L)}$, where the cross-entropy loss is applied directly through W_U . Earlier-layer gradients are obtained by backpropagating through subsequent attention, MLP, and normalization layers, picking up additional Jacobians along the way:

$$\nabla_{\mathbf{h}^{(e)}} \mathcal{L} = J_{\ell \rightarrow L}(\mathbf{h}^{(\ell)})^\top W_U^\top (\mathbf{p} - \mathbf{e}_y).$$

The fixed-direction pressure is therefore a property of the output-facing path. Earlier layers may maintain intermediate features in different geometric arrangements; for those features to affect the next-token distribution, however, they must eventually be converted into output-facing components readable by the unembedding.

Remark A.8 (From training pressure to learned representations). Proposition 3.1 characterizes the gradient *pressure* on $\mathbf{h}^{(L)}$, not the learned representation directly. The standard implicit assumption of any post-hoc analysis of trained networks is that representations have, after sufficient training, accumulated and aligned with the directions of repeated gradient pressure. Under this assumption, output-relevant features are exposed through the contrast directions $\mathbf{u}_y - \mathbf{u}_j$ that govern token decisions, and Theorem 3.3 below characterizes when projection onto these directions recovers the underlying feature magnitudes.

Remark A.9 (Why deeper readouts break independent-direction pressure). For a readout with $L_{\text{out}} > 2$ layers, the Jacobian becomes a chain product

$$J_G(\mathbf{h}) = W_{L_{\text{out}}} \prod_{l < L_{\text{out}}} \text{diag}(\sigma'_l) W_l,$$

in which each $\text{diag}(\sigma'_l)$ depends on the activations at layer l , which themselves depend on earlier gates. The effective gradient direction at any context is no longer a per-context-gated sum over a fixed set of column directions; it is a context-dependent linear combination *of* such sums. Feature pressures become coupled across readout layers, and the selection-only structure preserved by 2-layer MLPs no longer holds. This matches the empirical observation in Section 5.1: under linear and 2-layer MLP readouts, linear (Fourier) structure appears reliably across seeds, while under deeper nonlinear readouts it appears only intermittently.

Why shared directions are an efficient solution (full version). Propositions 3.1 and 3.2 establish what is and is not architecturally required. They do not by themselves explain why training tends to produce shared, reusable feature directions across many contexts. The following proposition fills that gap under an efficiency bias: among all representations satisfying a margin requirement through a linear readout, the minimum-norm solution is the one in which all such contexts share a single direction with magnitudes scaled by the required margin.

Proposition A.10 (Norm-Efficient Solutions Collapse onto Fixed Contrast Directions). *Fix a contrast direction $\mathbf{v} := \mathbf{u}_y - \mathbf{u}_j$, and let $\{c_i\}_{i=1}^N$ be contexts each requiring margin $m_i > 0$ on the (y, j)*

contrast. For each context, the minimum-norm representation that achieves the required margin under the linear readout,

$$\mathbf{h}_i^* = \arg\min_{\mathbf{h} \in \mathbb{R}^d} \|\mathbf{h}\|^2 \quad \text{subject to} \quad \mathbf{v}^\top \mathbf{h} \geq m_i,$$

is

$$\mathbf{h}_i^* = \frac{m_i}{\|\mathbf{v}\|^2} \mathbf{v}.$$

All $\mathbf{h}_i^*$ lie on the line spanned by $\mathbf{v}$, with magnitudes proportional to the required margin: same direction, context-dependent magnitude.

Proof. Decompose $\mathbf{h} = a\mathbf{v} + \mathbf{r}$ with $\mathbf{r} \perp \mathbf{v}$. The constraint $\mathbf{v}^\top \mathbf{h} = a\|\mathbf{v}\|^2 \geq m_i$ requires $a \geq m_i/\|\mathbf{v}\|^2$, and the norm $\|\mathbf{h}\|^2 = a^2\|\mathbf{v}\|^2 + \|\mathbf{r}\|^2$ is minimized at $\mathbf{r} = 0$ and $a = m_i/\|\mathbf{v}\|^2$. $\square$

Remark A.11 (Nonlinear readouts do not generally collapse). For a nonlinear margin $m_{y,j}(\mathbf{h}) = G_y(\mathbf{h}) - G_j(\mathbf{h})$, the corresponding minimum-norm problem replaces the linear constraint with $m_{y,j}(\mathbf{h}_i) \geq m_i$. At an active KKT point, $\mathbf{h}_i^*$ is parallel to $\nabla_{\mathbf{h}} m_{y,j}(\mathbf{h}_i^*)$, which depends on $\mathbf{h}_i^*$ for nonlinear G . Different contexts can therefore have different optimal directions; collapse to one shared direction occurs only under additional structure (e.g., affine $m_{y,j}$, or parallel margin gradients on the relevant domain). The non-collapse here parallels the non-necessity of Proposition 3.2.

Scope of the efficiency claim. Proposition A.10 characterizes the optimum of an explicit norm-minimization problem with one contrast direction. Real transformers do not solve this problem exactly, but several standard mechanisms produce a similar efficiency bias: weight decay penalizes parameter norms, gradient descent on classification objectives has a well-documented implicit bias toward small-norm solutions, and LayerNorm bounds activation norms at every block. The proposition formalizes what such an efficiency bias *prefers* when applied through a linear readout: contexts requiring the same contrast collapse onto a shared direction. With many contrast directions simultaneously, this composes into the analysis the paper develops throughout: each output-relevant feature occupies a fixed direction (or low-dimensional subspace) shared across the contexts that use it. The grokking ablation in Section 5.1 provides empirical support: under linear readouts, this directional structure emerges consistently across seeds; under deeper nonlinear readouts—where Remark A.11 says no such collapse is required—the structure emerges only intermittently. We do not claim this proposition derives feature emergence; emergence depends on optimization dynamics, training data, and inductive biases that existing work studies. The proposition characterizes the geometric *form* that an efficient solution takes when one is found.

Assumption A.12 (Sparse Feature Dictionary). Let $\mathcal{D} = \{\mathbf{d}_g\}_{g \in \mathcal{F}} \subset \mathbb{R}^d$ be a dictionary of unit-norm feature directions, $\|\mathbf{d}_g\| = 1$, with coherence

$$\mu := \max_{g \neq g'} |\langle \mathbf{d}_g, \mathbf{d}_{g'} \rangle|.$$

Assume that each context c admits a sparse feature decomposition

$$\mathbf{h}(c) = \sum_{g \in S(c)} \alpha_g(c) \mathbf{d}_g + \boldsymbol{\xi}(c),$$

where $S(c) \subseteq \mathcal{F}$ is the set of active features, $|S(c)| \leq k$, and $\|\boldsymbol{\xi}(c)\| \leq \varepsilon$. We say that context c expresses feature f when $f \in S(c)$ and $\alpha_f(c) \neq 0$.

This assumption formalizes the superposition hypothesis [Elhage et al., 2022]: features are encoded along an overcomplete dictionary of directions, and only a small number are active in any given context. We use $\mathbf{d}_f$ for the feature direction in the sparse-superposition analysis below; the bounds characterize when projection onto this direction is reliable. Whether $\mathbf{d}_f$ coincides with the architectural readout direction $\mathbf{w}_f$ from Proposition 3.1 is a separate identifiability question—in non-orthogonal dictionaries, the optimal linear readout for a feature can be closer to a dual vector than to the dictionary atom. We address this gap explicitly with a bounded-misalignment assumption.

Assumption A.13 (Readout–Dictionary Alignment). For each architecturally communicable feature f that also appears in the dictionary $\mathcal{D}$, the architectural readout direction $\mathbf{w}_f$ from Proposition 3.1 and the dictionary direction $\mathbf{d}_f$ from Assumption A.12 satisfy

$$\frac{\langle \mathbf{w}_f, \mathbf{d}_f \rangle}{\|\mathbf{w}_f\| \|\mathbf{d}_f\|} \geq 1 - \gamma$$

for some *readout misalignment* parameter $\gamma \in [0, 1)$. Equivalently, the angle between the two directions is at most $\arccos(1 - \gamma)$.

The scalar $\alpha_f(c)$ in the dictionary decomposition is, up to a multiplicative constant absorbed into the dictionary normalization, the same quantity as the readout output $g(f(c))$ from the proposition. The Theorem below is stated in terms of projections onto $\mathbf{d}_f$ for clarity; the corollary that follows shows projections onto $\mathbf{w}_f$ recover the same quantity up to an additional $O(\sqrt{\gamma})$ term.

Corollary A.14 (Readout–Dictionary Bridge). *Under Assumptions A.12 and A.13, projection onto the architectural readout direction $\mathbf{w}_f$ recovers the same feature magnitude as projection onto the dictionary direction $\mathbf{d}_f$, up to a misalignment-controlled additional term:*

$$\left| \frac{\langle \mathbf{h}(c), \mathbf{w}_f \rangle}{\|\mathbf{w}_f\|} - \langle \mathbf{h}(c), \mathbf{d}_f \rangle \right| \leq \sqrt{2\gamma} \|\mathbf{h}(c)\|.$$

Combined with Theorem 3.3, this gives

$$\left| \frac{\langle \mathbf{h}(c), \mathbf{w}_f \rangle}{\|\mathbf{w}_f\|} - \alpha_f(c) \right| \leq \mu(k-1)A + \varepsilon + \sqrt{2\gamma} \|\mathbf{h}(c)\|.$$

Proof. Let $\hat{\mathbf{w}}_f := \mathbf{w}_f / \|\mathbf{w}_f\|$ be the unit readout direction. By Assumption A.13, $\langle \hat{\mathbf{w}}_f, \mathbf{d}_f \rangle \geq 1 - \gamma$, so $\|\hat{\mathbf{w}}_f - \mathbf{d}_f\|^2 = 2 - 2\langle \hat{\mathbf{w}}_f, \mathbf{d}_f \rangle \leq 2\gamma$ (using $\|\hat{\mathbf{w}}_f\| = \|\mathbf{d}_f\| = 1$). Therefore by Cauchy–Schwarz,

$$|\langle \mathbf{h}(c), \hat{\mathbf{w}}_f \rangle - \langle \mathbf{h}(c), \mathbf{d}_f \rangle| = |\langle \mathbf{h}(c), \hat{\mathbf{w}}_f - \mathbf{d}_f \rangle| \leq \|\mathbf{h}(c)\| \|\hat{\mathbf{w}}_f - \mathbf{d}_f\| \leq \sqrt{2\gamma} \|\mathbf{h}(c)\|.$$

The combined bound follows by triangle inequality with Theorem 3.3. $\square$

Corollary A.15 (Projection-Based Detection). *Under the assumptions of Theorem 3.3, define the interference radius*

$$\rho(c) := \mu \sum_{g \in S(c), g \neq f} |\alpha_g(c)| + \varepsilon.$$

If c expresses f with $|\alpha_f(c)| > \rho(c)$, then $|\langle \mathbf{h}(c), \mathbf{d}_f \rangle| > 0$. Moreover, if non-expressing contexts satisfy $\alpha_f(c) = 0$, then a threshold $\tau > \rho(c)$ separates contexts with sufficiently large f -magnitude from contexts in which f is absent.

Proof. Theorem 3.3 gives $\langle \mathbf{h}(c), \mathbf{d}_f \rangle = \alpha_f(c) + e_f(c)$ with $|e_f(c)| \leq \rho(c)$. If $|\alpha_f(c)| > \rho(c)$, the projection cannot vanish. The threshold statement follows from the same bound. $\square$

Corollary A.16 (Margin Condition for Multiclass Classification). *Consider a K -class task represented as one-vs-rest class indicators with directions $\{\mathbf{d}_1, \dots, \mathbf{d}_K\}$. Let c have true class y , and define the projection classifier*

$$\hat{y}(c) = \arg \max_{j \in [K]} \langle \mathbf{h}(c), \mathbf{d}_j \rangle.$$

Suppose each class projection has interference at most ρ , i.e., $|\langle \mathbf{h}(c), \mathbf{d}_j \rangle - \alpha_j(c)| \leq \rho$ for all $j \in [K]$. If the true class-feature margin satisfies $\alpha_y(c) - \max_{j \neq y} \alpha_j(c) > 2\rho$, then $\hat{y}(c) = y$.

Proof. For the true class y : $\langle \mathbf{h}(c), \mathbf{d}_y \rangle \geq \alpha_y(c) - \rho$. For any incorrect class $j \neq y$: $\langle \mathbf{h}(c), \mathbf{d}_j \rangle \leq \alpha_j(c) + \rho \leq \max_{j' \neq y} \alpha_{j'}(c) + \rho$. The score for y exceeds that of every other class whenever $\alpha_y(c) - \rho > \max_{j \neq y} \alpha_j(c) + \rho$, which is the stated margin condition. $\square$

Assumption A.17 (Token Feature Bundles). *For the output tokens considered in a task, assume the unembedding row $\mathbf{u}_t = (W_U)_{t,:}$ admits an approximate sparse decomposition over the same feature dictionary $\mathcal{D} = \{\mathbf{d}_f\}_{f \in \mathcal{F}}$ used to decompose residual-stream activations in Assumption A.12:*

$$\mathbf{u}_t = \sum_{f \in B(t)} b_{t,f} \mathbf{d}_f + \mathbf{r}_t.$$

Here $B(t) \subseteq \mathcal{F}$ is the token’s feature bundle, $b_{t,f} \in \mathbb{R}$ is the contribution of feature f to token t , and $\mathbf{r}_t \in \mathbb{R}^d$ is the unmodeled residual with $\|\mathbf{r}_t\| \leq \rho_t$. The bundle support sizes $|B(t)|$ are assumed sparse on the same order as the residual-stream support sizes $|S(c)|$.

This assumption is consistent with empirical findings on superposition in transformer matrices [Elhage et al., 2022]: just as the residual stream is modeled as a sparse combination of dictionary atoms, the unembedding rows are expected to be sparse combinations of the same atoms. We do not require this decomposition to hold for every token in the vocabulary, only for those participating in the task at hand.

The bundle-matching proposition gives a concrete architectural interpretation of next-token prediction: the unembedding scores each candidate token by how well its feature bundle matches the active features currently encoded in the residual stream. This sharpens the architectural argument: every token whose probability the model can affect must have its bundle features exposed in directions the unembedding can read, because the unembedding has no other way to score them. The Self-Reference Property below follows directly: a class prompt’s unembedding row is itself a feature bundle for that class, so its activation aligns with the same dictionary directions as any other context expressing those features.

Corollary A.18 (Feature Contributions to Logit Margins). *Let y and j be two output tokens with unembedding rows $\mathbf{u}_y$ and $\mathbf{u}_j$, and define the contrast direction*

$$\mathbf{v}_{y,j} := \mathbf{u}_y - \mathbf{u}_j.$$

Suppose $\mathbf{v}_{y,j}$ decomposes over the feature dictionary as

$$\mathbf{v}_{y,j} = \sum_{f \in B(y,j)} \beta_{y,j,f} \mathbf{d}_f + \mathbf{r}_{y,j},$$

with $\|\mathbf{r}_{y,j}\| \leq \rho_{y,j}$. Then for a context activation $\mathbf{h}(c) = \sum_{g \in S(c)} \alpha_g(c) \mathbf{d}_g + \boldsymbol{\xi}(c)$ as in Assumption A.12, the logit margin satisfies

$$\ell_y(c) - \ell_j(c) = \sum_{f \in B(y,j) \cap S(c)} \beta_{y,j,f} \alpha_f(c) + \Delta_{y,j}(c),$$

where the interference term satisfies

$$|\Delta_{y,j}(c)| \leq \mu \sum_{f \in B(y,j)} |\beta_{y,j,f}| \sum_{\substack{g \in S(c) \\ g \neq f}} |\alpha_g(c)| + \|\mathbf{r}_{y,j}\| \|\mathbf{h}(c)\| + \|\mathbf{v}_{y,j}\| \|\boldsymbol{\xi}(c)\|.$$

A feature f supports token y over j when $\beta_{y,j,f} > 0$ and opposes it when $\beta_{y,j,f} < 0$. Thus, features contribute to the model’s token choice through fixed output-facing contrast directions, up to bounded sparse-superposition interference.

Proof. By linearity, $\ell_y(c) - \ell_j(c) = (\mathbf{u}_y - \mathbf{u}_j)^\top \mathbf{h}(c) = \mathbf{v}_{y,j}^\top \mathbf{h}(c)$. Applying Proposition A.1 with the contrast direction $\mathbf{v}_{y,j}$ in place of a single unembedding row $\mathbf{u}_t$ yields the matching term and the stated interference bound; the bundle support $B(y,j)$ collects all features that distinguish the two tokens, and $\mathbf{r}_{y,j}$ is the unmodeled component of the contrast. $\square$

This corollary connects the framework directly to model decisions: classification between alternatives is governed by feature-weighted contrast scores, with each feature’s contribution determined by how it appears in the relevant unembedding contrast direction. It also explains a class of empirically observed phenomena: features that distinguish closely related tokens (e.g., singular vs. plural, masculine vs. feminine, member-of-class vs. not-member) should appear with opposing signs in $\mathbf{v}_{y,j}$, and the magnitude of $\beta_{y,j,f}$ predicts the strength of that feature’s effect on token choice.

Remark A.19 (Multiclass features). For multiclass datasets, we do not assume that the entire K -valued class variable occupies a single one-dimensional subspace. Instead, we treat the task as a collection of one-vs-rest class indicators. The full class variable may occupy a higher-dimensional class subspace (Proposition C.1 in the appendix shows $\dim(\mathcal{S}_f) \leq K - 1$), while each class indicator is associated with its own directional readout. This is the setting used by the projection classifier in Corollary A.16 and by the class-token probes in Section 4.

Remark A.20 (Compatibility with the $\alpha_f \mathbf{d}_f + \boldsymbol{\eta}$ notation). Throughout the rest of the paper we sometimes write the decomposition compactly as $\mathbf{h}(c) = \alpha_f(c) \mathbf{d}_f + \boldsymbol{\eta}(c)$, where $\boldsymbol{\eta}(c) := \sum_{g \in S(c), g \neq f} \alpha_g(c) \mathbf{d}_g + \boldsymbol{\xi}(c)$ collects all non- f contributions. We do not require $\boldsymbol{\eta}(c) \perp \mathbf{d}_f$: under superposition, $\boldsymbol{\eta}(c)$ generally has nonzero overlap with $\mathbf{d}_f$, and Theorem 3.3 bounds exactly this overlap.

Note on basis-dependence. Residual stream geometry is non-canonical: applying any invertible $A \in \mathbb{R}^{d \times d}$ to the residual stream and absorbing A^{-1} into all weight matrices yields a behaviorally equivalent model with different-looking “directions” in coordinates. The invariant direction $\mathbf{d}_f$ is therefore not a basis-independent geometric object in absolute terms; Park et al. [2023] address this concern formally via causal inner products. The framework’s claim is more restricted but, we argue, sufficient for its purpose: *within a fixed parameterization, all activations and weight matrices share the same basis*. Class-prompt activations $\mathbf{h}_t$ and instance activations $\mathbf{h}(c)$ both inhabit the same residual-stream coordinates as the model that produced them, so projecting one onto the other is well-defined regardless of basis choice. If A transforms the residual stream, both $\mathbf{h}_t$ and $\mathbf{h}(c)$ transform together, and the projection $\mathbf{h}_t^\top \mathbf{h}(c)$ is invariant up to the metric implied by A . Identifiability concerns become substantive only when comparing directions *across* parameterizations (e.g., across models, training runs, or analyses that fix one space and vary another); they do not affect the framework’s intended use of within-model self-reference.

A.5 Capacity Constraints and Feature Sharing

The architectural argument so far identifies fixed decoding directions for features that are linearly communicable through some interface in $\mathcal{L}$. A complementary question is why features should be *factorized* into a small number of shared directions in the first place, rather than each token occupying its own dedicated direction. The vocabulary size $|\mathcal{V}|$ greatly exceeds the residual stream dimension d in practical transformers, which means most tokens must share structure with others. Under standard assumptions about feature sparsity and co-occurrence, this capacity constraint favors factorized representations: a small set of shared feature directions, with each token expressed as a sparse combination. We give a detailed argument in Appendix C; the conclusion connects directly to the superposition hypothesis [Elhage et al., 2022], with each shared factor satisfying the conditions of Theorem 3.3.

Remark A.21 (Implicit Classification). From this view, predicting a token *is* implicit classification: the model checks “does this context encode the factor combination for token t ?” Each factor $\mathbf{d}_f$ partitions contexts into those expressing f versus those that do not. Compression pressure from the capacity constraint ensures factors are reusable across tokens, maximizing the number of features that can coexist in the representation, which aligns with the information bottleneck principle [Tishby and Zaslavsky, 2015].

The decomposition $\mathbf{h}(c) = \alpha_f(c) \mathbf{d}_f + \boldsymbol{\eta}(c)$ from Theorem 3.3 is central to the framework: the *direction* $\mathbf{d}_f$ is invariant across contexts, while the *magnitude* $\alpha_f(c)$ varies. Linear interfaces preserve this structure:

$$W(\alpha \cdot \mathbf{d}_f) = \alpha \cdot (W\mathbf{d}_f) \quad (8)$$

A natural concern is whether LayerNorm—an input-dependent normalization applied before each block—disrupts directional invariance. It does not, in the following sense:

Proposition A.22 (LayerNorm preserves directional invariance). *Let $\mathbf{h}(c) = \alpha_f(c) \mathbf{d}_f + \boldsymbol{\eta}(c)$ where $\mathbf{d}_f \in \mathbb{R}^d$ is a context-invariant feature direction and $\alpha_f(c) \in \mathbb{R}$ is a context-dependent magnitude. If $\mathbf{d}_f \notin \text{span}(\mathbb{1})$, then there exists a context-invariant direction $\tilde{\mathbf{d}}_f$ such that*

$$\text{LN}(\mathbf{h}(c)) = \tilde{\alpha}_f(c) \tilde{\mathbf{d}}_f + \tilde{\boldsymbol{\eta}}(c) + \boldsymbol{\beta},$$

where $\tilde{\alpha}_f(c) := \alpha_f(c)/\sigma(\mathbf{h}(c))$ and $\boldsymbol{\beta}$ is the learned LayerNorm bias.

LayerNorm maps a context-invariant direction $\mathbf{d}_f$ to a transformed context-invariant direction $\tilde{\mathbf{d}}_f = \gamma \odot \Pi_{\mathbb{1}^\perp} \mathbf{d}_f$, where γ is the learned scale and $\Pi_{\mathbb{1}^\perp}$ is the mean-centering projection. The transformation depends only on $\mathbf{d}_f$ and the learned LayerNorm parameters; context enters only through the scalar magnitude $\tilde{\alpha}_f(c)$. Directions remain context-invariant; magnitudes are rescaled by $1/\sigma(\mathbf{h}(c))$. Full proof in Appendix B.

A.6 The Identity-Projection Operator

We now introduce the operator that enables practical applications, and use it to make the framework’s causal claim explicit.

Definition A.23 (Identity-Projection Operator). For a context c and feature f with invariant direction $\mathbf{d}_f$, the *Identity-Projection* is:

$$\mathcal{I}_f(c) \triangleq \mathbf{h}(c)^\top \hat{\mathbf{d}}_f \quad (9)$$

where $\hat{\mathbf{d}}_f = \mathbf{d}_f / \|\mathbf{d}_f\|$ is the unit direction.

The Identity-Projection operator measures the magnitude of feature f within the representation $\mathbf{h}(c)$. By Theorem 3.3, f flows through $\mathbf{h}(c)$ along the fixed direction $\mathbf{d}_f$ with magnitude $\alpha_f(c)$; the projection onto $\mathbf{d}_f$ recovers this magnitude up to the sparse-superposition interference term $\rho(c)$ from Corollary A.15.

Linear-path contribution. The magnitude $\alpha_f(c)$ is not merely a descriptive quantity. Because the architecture reads features through linear interfaces (Definition A.4), $\alpha_f(c)$ is also *the quantity that flows downstream through linear paths*: an interface $L \in \mathcal{L}$ reading the residual stream produces $L\mathbf{h}(c) = \alpha_f(c)L\mathbf{d}_f + L\boldsymbol{\eta}(c)$, so within linear downstream paths, the contribution of f to whatever L computes is proportional to $\alpha_f(c)$. The framework therefore makes a quantitative *linear-path* contribution claim: $\alpha_f(c)$ measures how much f contributes to downstream behavior *when the path from $\mathbf{h}(c)$ to the output is linear*. A feature with $\alpha_f(c) = 0$ is inert through linear paths at c ; a feature with large $\alpha_f(c)$ drives downstream linear computations in proportion. This linear-path account is consistent with the qualitative pattern reported in activation patching studies—intervening on the $\mathbf{d}_f$ -component of $\mathbf{h}(c)$ tends to produce downstream effects scaled by the change in α_f —though we do not run intervention experiments in this paper, so the full causal claim is left to future work.

Proposition A.24 (Feature Operations). *For a feature f satisfying the conditions of Theorem 3.3:*

- (i) **Detection:** $\mathcal{I}_f(c) > \tau$ indicates f is expressed in c above threshold τ .
- (ii) **Linear-path magnitude:** $\mathcal{I}_f(c)$ measures the magnitude of f in c , which equals (up to scaling by $\|\mathbf{d}_f\|$) the magnitude that flows through any downstream linear interface that reads from $\mathbf{h}(c)$.

Proof. From the decomposition $\mathbf{h}(c) = \alpha_f(c)\mathbf{d}_f + \boldsymbol{\eta}(c)$ in Theorem 3.3:

$$\mathcal{I}_f(c) = \alpha_f(c)\|\mathbf{d}_f\| + \boldsymbol{\eta}(c)^\top \hat{\mathbf{d}}_f. \quad (10)$$

The second term reflects superposition—components of $\boldsymbol{\eta}(c)$ corresponding to other features may have nonzero overlap with $\hat{\mathbf{d}}_f$, since transformer features are not required to occupy orthogonal directions [Elhage et al., 2022]. In practice this overlap is bounded by the sparsity of co-active features and acts as a noise floor on $\mathcal{I}_f$; the central content of (i) and (ii) is that the leading term $\alpha_f(c)\|\mathbf{d}_f\|$ tracks the magnitude of f both as it appears in the representation and as it flows downstream through linear interfaces. $\square$

A.7 Extension to MLP Blocks via Selection-Only Decomposition

Proposition 3.1 applies to features decoded through fixed linear interfaces. A natural concern is whether the framework extends to features that are written by MLP blocks, which contain a per-coordinate nonlinearity. We show that under a structural condition—which we call a *selection-only feature decomposition*—MLP-written features inherit the framework’s invariant-direction conclusions. We then show that 2-layer MLPs structurally satisfy this condition, while deeper MLPs admit a write-direction decomposition but break the independence of feature magnitudes through compositional gating.

Let an MLP block be

$$\text{MLP}(\mathbf{x}) = W_{\text{out}} \cdot \sigma(W_{\text{in}}\mathbf{x} + \mathbf{b}_{\text{in}}) + \mathbf{b}_{\text{out}},$$

with $W_{\text{in}} \in \mathbb{R}^{m \times d}$, $W_{\text{out}} \in \mathbb{R}^{d \times m}$, and σ a per-coordinate nonlinearity. Denote the i -th row of W_{in} by $\mathbf{w}_i^{\text{in}} \in \mathbb{R}^d$ (read direction) and the i -th column of W_{out} by $\mathbf{w}_i^{\text{out}} \in \mathbb{R}^d$ (write direction).

Definition A.25 (Selection-Only Feature Decomposition). An MLP block admits a *selection-only feature decomposition* if there exists a finite collection of feature directions $\{\mathbf{d}_f\}_{f \in F}$ in the residual stream and scalar functions $\{m_f(\mathbf{x})\}_{f \in F}$ such that

$$\text{MLP}(\mathbf{x}) = \sum_{f \in F} m_f(\mathbf{x}) \cdot \mathbf{d}_f + \mathbf{b}_{\text{out}}$$

where:

- (a) **Context-independence of directions.** Each direction $\mathbf{d}_f \in \mathbb{R}^d$ is fixed and does not depend on $\mathbf{x}$. The input controls only the scalar magnitude $m_f(\mathbf{x})$ associated with each direction; it does not change the direction itself.
- (b) **Independent magnitudes.** The contribution along each $\mathbf{d}_f$ is controlled by its own scalar $m_f(\mathbf{x})$, and the value of $m_f(\mathbf{x})$ does not depend on the magnitudes $\{m_{f'}(\mathbf{x})\}_{f' \neq f}$ of other features.

The central claim is condition (a): *directions are context-independent; only magnitudes vary with input*. Whether $m_f(\mathbf{x})$ is large, small, zero, or negative in any given context is not constrained—what is constrained is that the direction $\mathbf{d}_f$ along which f ’s contribution is written remains the same across all contexts. Condition (b) is a structural requirement we use below to distinguish 2-layer from deeper MLPs: each feature’s magnitude is determined independently of other features’ magnitudes, so no feature’s contribution is coupled to the gating of others.

Proposition A.26 (MLP Extension of Invariant Subspace Structure). *Suppose an MLP block admits a selection-only feature decomposition in the sense of Definition A.25. Then for each feature f in the decomposition:*

1. **(Identification)** $\mathbf{d}_f$ is a fixed direction in the residual stream, identifiable from the MLP weights.
2. **(Cross-context invariance)** $\mathbf{d}_f$ does not depend on the input; only $m_f(\mathbf{x})$ does.
3. **(Pathway compatibility)** f is decodable by any downstream linear interface $L \in \mathcal{L}$ whose row space includes $\mathbf{d}_f$, with the decoded value proportional to $m_f(\mathbf{x})$.

The conclusions of Proposition 3.1 thus extend to MLP-written features under the selection-only condition.

Proof sketch. Direct from Definition A.25: (1) and (2) restate condition (a). For (3), any linear functional L applied to $\sum_f m_f(\mathbf{x})\mathbf{d}_f$ yields $\sum_f m_f(\mathbf{x}) \cdot (L\mathbf{d}_f)$, preserving the selection-only structure with modified scalar coefficients. $\square$

Proposition A.27 (Architectural Admissibility of Selection-Only Decomposition). *A 2-layer MLP $\text{MLP}(\mathbf{h}) = W_{\text{out}}\sigma(W_{\text{in}}\mathbf{h} + \mathbf{b}_{\text{in}}) + \mathbf{b}_{\text{out}}$ structurally satisfies the selection-only condition (Definition A.25): the columns of W_{out} are the fixed write directions, and the gates $g_i(\mathbf{h}) = \sigma(\langle \mathbf{w}_i^{\text{in}}, \mathbf{h} \rangle + b_{\text{in},i})$ depend only on each unit’s own read direction. For an L_{out} -layer MLP with $L_{\text{out}} > 2$, the output retains fixed final write directions but the gates depend on compositions of upstream activations, breaking the independent-magnitude condition. The framework’s invariant-direction conclusions therefore extend to features written by 2-layer MLP blocks; deeper MLPs preserve write-direction structure but couple feature magnitudes through compositional gating. The proof appears in Appendix B.*

Remark A.28 (Empirical Realization of the Dictionary Condition). Whether a trained 2-layer MLP exhibits a selection-only decomposition aligned with semantically meaningful features is an empirical question. The architecture satisfies the structural conditions (Proposition A.27); whether training realizes an interpretable dictionary within this structure requires evidence. Three lines of existing work support this for trained transformers: (i) sparse autoencoder findings showing that MLP activations decompose into sparse, monosemantic features with stable directions [Bricken et al., 2023, Cunningham et al., 2023]; (ii) linear probing across layers showing that class-feature directions remain linearly recoverable through MLP processing [Park et al., 2023]; and (iii) circuit tracing methods such as edge attribution patching, which decompose MLP contributions into per-component additive effects—a decomposition that presupposes selection-only structure. The framework’s structure is therefore: *architecture provides the structural conditions* (Proposition A.27, with the qualitative break at $L > 2$ in condition (b)), while *training provides realization* (cited evidence above).

A.8 The Self-Reference Property

A key practical question is how to obtain a usable reference for $\mathbf{d}_f$ without labeled instance data or direct access to the relevant interface. Class prompts provide such a reference because they are

themselves contexts expressing the feature. However, their activations need not be pure: a class prompt may also contain lexical, syntactic, frequency, or category-level components. We therefore state self-reference as an approximate property whose quality depends on the size of the target feature component relative to the residual.

Corollary A.29 (Approximate Self-Reference). *Let t_f be a token or expression that canonically expresses feature f . Assume its activation decomposes as*

$$\mathbf{h}_{t_f} = \beta_f \mathbf{d}_f + \mathbf{u}_f,$$

where $\|\mathbf{d}_f\| = 1$, $\beta_f > 0$, and $\|\mathbf{u}_f\| \leq \delta_f$. If $\delta_f < \beta_f$, then the normalized class-prompt activation

$$\hat{\mathbf{h}}_{t_f} := \frac{\mathbf{h}_{t_f}}{\|\mathbf{h}_{t_f}\|}$$

has positive alignment with the true feature direction. More specifically,

$$\cos(\hat{\mathbf{h}}_{t_f}, \mathbf{d}_f) = \langle \hat{\mathbf{h}}_{t_f}, \mathbf{d}_f \rangle \geq \frac{\beta_f - \delta_f}{\beta_f + \delta_f}.$$

Thus, a class prompt is a good empirical reference for $\mathbf{d}_f$ when its target feature component dominates its residual components.

Proof. We have $\langle \mathbf{h}_{t_f}, \mathbf{d}_f \rangle = \beta_f + \langle \mathbf{u}_f, \mathbf{d}_f \rangle \geq \beta_f - \|\mathbf{u}_f\| \geq \beta_f - \delta_f$. Also, $\|\mathbf{h}_{t_f}\| = \|\beta_f \mathbf{d}_f + \mathbf{u}_f\| \leq \beta_f + \|\mathbf{u}_f\| \leq \beta_f + \delta_f$. Therefore $\cos(\hat{\mathbf{h}}_{t_f}, \mathbf{d}_f) = \langle \mathbf{h}_{t_f}, \mathbf{d}_f \rangle / \|\mathbf{h}_{t_f}\| \geq (\beta_f - \delta_f) / (\beta_f + \delta_f)$. $\square$

Each term in the margin condition maps to a measurable quantity: $\alpha_y(c)$ is the true class signal strength, μ_C is cross-class direction coherence, ρ_j is instance interference along class j ’s direction, and δ_j is class-prompt contamination. Failure modes correspond to violations of specific terms: low α_y (weak signal), high μ_C (overlapping class directions), high ρ_j (instance contains active features along other classes’ directions), or high δ_j (class prompts encode substantial off-feature content).

Remark A.30 (Class prompts as approximate references). The practical implication is that running a class prompt such as “France” through the model can provide a reference direction for the France feature, but only approximately. The class-prompt activation may also encode lexical identity, token frequency, broad category information such as “being a country,” and other contextual components. Mean-centering class prompts, as in Eq. 1, can reduce shared category-level components, while learned unsupervised transformations can further improve separability.

The framework’s name reflects this structure: the model itself, queried with a canonical class prompt, returns an approximate reference for the corresponding feature direction in a single forward pass. The reliability of this reference depends empirically on how strongly the target-feature signal dominates other components of the prompt’s activation.

The mechanism behind Self-Reference: feature aggregation and grouping pressure (full version).

The Self-Reference Property is most natural when explained mechanistically: the same feature dictionary $\{\mathbf{d}_f\}$ that organizes residual activations also organizes how attention writes back to the prediction position and how the unembedding scores candidates. In the regime where attention’s OV map approximately respects the dictionary—a mechanism compatible with the assumption that residual activations across positions are expressed in a shared feature dictionary, which is already implicit in our setup—attention aggregates feature evidence from prompt tokens into the prediction position, where the unembedding then matches it against next-token feature bundles. The following two results formalize this and the gradient-level pressure that reinforces it during training.

Proposition A.31 (Attention Aggregates Shared Feature Evidence). *Consider an attention head writing to prediction position q ,*

$$\mathbf{o}_q = W_O \sum_i a_{q,i} W_V \mathbf{h}_i,$$

with token-position activations decomposing over the shared dictionary $\mathbf{h}_i = \sum_f \alpha_{i,f} \mathbf{d}_f + \boldsymbol{\xi}_i$. Suppose the OV map $M := W_O W_V$ is locally feature-compatible for feature f on the contexts considered, in the sense that

$$M \mathbf{d}_f = \lambda_f \mathbf{d}_f + \mathbf{e}_f, \quad |\langle \mathbf{e}_f, \mathbf{d}_f \rangle| \leq \epsilon_f.$$

We do not require this condition to hold universally across all heads or features; it characterizes the regime in which attention preserves feature f for the head and contexts under analysis. Then the attention-written contribution along $\mathbf{d}_f$ is

$$\langle \mathbf{o}_q, \mathbf{d}_f \rangle = \lambda_f \sum_i a_{q,i} \alpha_{i,f} + E_f(q),$$

where $E_f(q)$ collects cross-feature and residual interference. When many attended tokens express feature f (large $\alpha_{i,f}$), the head’s output along $\mathbf{d}_f$ reflects their attention-weighted aggregate.

Proof. Substitute the decomposition of $\mathbf{h}_i$ into $\mathbf{o}_q$ and use linearity: $\mathbf{o}_q = \sum_i a_{q,i} M \mathbf{h}_i = \sum_i a_{q,i} (\sum_g \alpha_{i,g} (\lambda_g \mathbf{d}_g + \mathbf{e}_g) + M \boldsymbol{\xi}_i)$. Projecting onto $\mathbf{d}_f$ and isolating the $g = f$ matching term gives the stated identity, with $E_f(q)$ collecting the off-feature ($g \neq f$) contributions, the $\mathbf{e}_g$ residuals, and $\langle M \boldsymbol{\xi}_i, \mathbf{d}_f \rangle$. $\square$

Corollary A.32 (Cross-Entropy Reinforces Shared Feature Coordinates). *Let $\alpha_f(q)$ denote the residual-stream magnitude along $\mathbf{d}_f$ at the prediction position, and let unembedding rows decompose as feature bundles $\mathbf{u}_t = \sum_f b_{t,f} \mathbf{d}_f + \mathbf{r}_t$. Up to interference, the logit for token t contains the feature contribution $b_{t,f} \alpha_f(q)$. The cross-entropy gradient with respect to $\alpha_f(q)$ is*

$$\frac{\partial \mathcal{L}}{\partial \alpha_f(q)} = \sum_t p_t(c) b_{t,f} - b_{y,f} + (\text{interference}),$$

which is negative whenever $b_{y,f} > \sum_t p_t(c) b_{t,f}$ —i.e., when the correct next token contains feature f more strongly than the model’s current predicted distribution does. In that regime, gradient descent increases $\alpha_f(q)$.

Proof. The logit $\ell_t(c)$ contains the term $b_{t,f} \alpha_f(q)$ from $\mathbf{u}_t^\top \mathbf{h}(c)$ by Proposition A.1. Differentiating cross-entropy $\mathcal{L} = -\log p_y(c)$ through the softmax gives $\partial \mathcal{L} / \partial \ell_t = p_t(c) - \mathbf{1}[t = y]$, and the chain rule with $\partial \ell_t / \partial \alpha_f(q) = b_{t,f}$ (up to interference) yields the stated identity. $\square$

Together, Proposition A.31 and Corollary A.32 describe a closed loop: attention aggregates feature evidence from prompt tokens into the prediction position; the unembedding scores candidates by feature-bundle matching against this aggregated representation; cross-entropy then reinforces this coordination during training, increasing the shared feature magnitude whenever the target token contains the feature more strongly than the prediction does. Repeated across many contexts, this is a *grouping pressure*: surface forms that all express the same underlying feature are routed through the same output-facing coordinate because they all support next tokens whose unembedding bundles include the corresponding feature component. We do not claim this proves features must emerge—emergence still depends on optimization dynamics, training data, and inductive biases—but it provides the mechanistic content of *why* shared feature directions are useful solutions to next-token prediction once the architecture commits to linear or selection-only readouts. The connection to next-token prediction as implicit classification is made explicit in Remark A.21.

A.9 Empirical Validation of the Self-Reference Property

Corollary A.29 predicts that the activation $\mathbf{h}_t$ produced by a class prompt t should align with the activations $\mathbf{h}(c)$ produced by other contexts c expressing the same feature. We test this by comparing each class prompt’s hidden state against the mean instance hidden state for each class, computed across attention heads.

Specifically, for each attention head and class k , we compute the cosine similarity between the class token direction $\mathbf{h}_{t_k}$ and the mean instance direction $\bar{\mathbf{h}}_k$ for the same class (within-class) versus other classes (between-class). If tokens provide feature directions as predicted, within-class similarity should consistently exceed between-class similarity.

Figure 7 confirms this prediction: across 89.6%-98.6% of attention heads, the class token aligns more strongly with its own class instances than with any other class. This pattern validates the Self-Reference Property—tokens encode the same directional features present in contexts expressing those features, enabling zero-shot extraction of semantic directions.

The 2–14% of heads that fall below the diagonal are predicted by the framework rather than counterexamples to it. Each attention head encodes a particular subset of features (positional, syntactic, copying, semantic), and a class token aligns only with heads whose features it actually uses; heads that encode unrelated features show no alignment. A second source of off-diagonal points is *partial-synonym* structure: a feature shared by multiple classes (e.g., a feature common to “cat” and “dog”) will, in a head that encodes it, align both classes to the same direction. Instances remain more aligned with their own class overall, because the larger set of class-relevant heads—in both number and magnitude—dominates the aggregate, even when no individual head encodes the whole class.

B Proofs of Main Results

B.1 Proofs of Lemma A.1, Propositions 3.1 and A.1, and Theorem 3.3

Proof of Lemma A.1 (Linear-Path Communication Lemma).

Proof. Let $L \in \mathcal{L}$ be a fixed linear interface and $\phi \in \mathbb{R}^{k_L}$ a fixed downstream readout direction. The downstream scalar quantity is

$$z(c) = \phi^\top L \mathbf{h}(c) = (L^\top \phi)^\top \mathbf{h}(c) = \mathbf{w}^\top \mathbf{h}(c), \quad (11)$$

where $\mathbf{w} := L^\top \phi \in \mathbb{R}^d$. The direction $\mathbf{w}$ is fully determined by L (a fixed model parameter, e.g., W_U or an attention QKV matrix) and ϕ (a fixed readout direction in interface space, e.g., the indicator vector $\mathbf{e}_t$ for a token t). Since both are fixed at inference, $\mathbf{w}$ does not depend on c . No empirical estimation or supervised training is required to identify $\mathbf{w}$. $\square$

Proof of Proposition 3.1 (Linear Unembedding Imposes Fixed Contrast Coordinates).

Proof. The full proof is given in the main text. We restate it briefly. The inference identity $\ell_y(c) - \ell_j(c) = (\mathbf{u}_y - \mathbf{u}_j)^\top \mathbf{h}^{(L)}(c)$ follows directly from the linearity of the unembedding, since $\ell(c) = W_U \mathbf{h}^{(L)}(c)$ and $\mathbf{u}_t$ is the t -th row of W_U .

For the training-time identity, let $\ell(c) = W_U \mathbf{h}^{(L)}(c)$ and $\mathbf{p}(c) = \text{softmax}(\ell(c))$. The cross-entropy loss for target token y is $\mathcal{L}(c) = -\ell_y(c) + \log \sum_t \exp(\ell_t(c))$. Differentiating with respect to logits gives $\partial \mathcal{L} / \partial \ell_t = p_t(c) - \mathbf{1}[t = y]$, so $\nabla_{\ell} \mathcal{L} = \mathbf{p}(c) - \mathbf{e}_y$. The chain rule with $\ell = W_U \mathbf{h}^{(L)}$ then gives

$$\nabla_{\mathbf{h}^{(L)}} \mathcal{L}(c) = W_U^\top (\mathbf{p}(c) - \mathbf{e}_y) = \sum_{t \in \mathcal{V}} (p_t(c) - \mathbf{1}[t = y]) \mathbf{u}_t. \quad (12)$$

Since $\sum_t (p_t(c) - \mathbf{1}[t = y]) = 0$, the gradient rearranges as $\sum_{j \neq y} p_j(c) (\mathbf{u}_j - \mathbf{u}_y)$, a probability-weighted sum of contrast directions $\mathbf{u}_j - \mathbf{u}_y$. The same fixed contrast directions that govern inference therefore govern training-time updates as well. $\square$

Proof of Proposition A.1 (Nonlinear Readouts Relax Output-Facing Directional Pressure).

Proof. For a general readout $\ell(c) = G(\mathbf{h}^{(L)}(c))$, the chain rule gives $\nabla_{\mathbf{h}^{(L)}} \mathcal{L} = J_G(\mathbf{h}^{(L)})^\top (\mathbf{p}(c) - \mathbf{e}_y)$. The Jacobian rows define the directions through which the loss applies pressure to $\mathbf{h}^{(L)}$; when G is nonlinear, these depend on the activation and hence on c .

For the 2-layer MLP $G(\mathbf{h}) = W_{\text{out}} \sigma(W_{\text{in}} \mathbf{h} + \mathbf{b})$, differentiating layer by layer:

$$J_G(\mathbf{h}) = W_{\text{out}} \text{diag}(\sigma'(W_{\text{in}} \mathbf{h} + \mathbf{b})) W_{\text{in}}. \quad (13)$$

Therefore

$$\nabla_{\mathbf{h}^{(L)}} \mathcal{L} = W_{\text{in}}^\top \text{diag}(\sigma'(\cdot)) W_{\text{out}}^\top (\mathbf{p}(c) - \mathbf{e}_y), \quad (14)$$

which is a sum over the columns of $W_{\text{in}}^\top$ (the read directions of the MLP head), with weights determined by the per-context gates $\sigma'(\cdot)$ and the prediction-error vector. The columns of $W_{\text{in}}^\top$ are fixed; only their scalar weights depend on c . Forward writes through the head use the fixed columns of W_{out} ; backward pressure on $\mathbf{h}^{(L)}$ uses the fixed columns of $W_{\text{in}}^\top$. Selection-only structure is preserved in both directions. $\square$

Proof of Theorem 3.3 (Invariant Direction under Sparse Superposition).

Proof. The full proof is given in the main text following the theorem statement. We restate the steps for completeness. Let $\mathcal{D} = \{\mathbf{d}_g\}_{g \in \mathcal{F}}$ be a unit-norm dictionary with coherence μ , and assume $\mathbf{h}(c) = \sum_{g \in S(c)} \alpha_g(c) \mathbf{d}_g + \boldsymbol{\xi}(c)$ with $|S(c)| \leq k$ and $\|\boldsymbol{\xi}(c)\| \leq \varepsilon$.

Separating the f term (taking $\alpha_f(c) = 0$ if $f \notin S(c)$),

$$\mathbf{h}(c) = \alpha_f(c) \mathbf{d}_f + \sum_{g \in S(c), g \neq f} \alpha_g(c) \mathbf{d}_g + \boldsymbol{\xi}(c). \quad (15)$$

Projecting onto $\mathbf{d}_f$ with $\|\mathbf{d}_f\| = 1$,

$$\langle \mathbf{h}(c), \mathbf{d}_f \rangle = \alpha_f(c) + \sum_{g \in S(c), g \neq f} \alpha_g(c) \langle \mathbf{d}_g, \mathbf{d}_f \rangle + \langle \boldsymbol{\xi}(c), \mathbf{d}_f \rangle. \quad (16)$$

By the triangle inequality, the coherence bound $|\langle \mathbf{d}_g, \mathbf{d}_f \rangle| \leq \mu$, and Cauchy-Schwarz with $\|\mathbf{d}_f\| = 1$ and $\|\boldsymbol{\xi}(c)\| \leq \varepsilon$,

$$|\langle \mathbf{h}(c), \mathbf{d}_f \rangle - \alpha_f(c)| \leq \mu \sum_{g \in S(c), g \neq f} |\alpha_g(c)| + \varepsilon. \quad (17)$$

The simplified bound $\mu(k-1)A + \varepsilon$ follows from $|S(c) \setminus \{f\}| \leq k-1$ and the assumption $|\alpha_g(c)| \leq A$ for all active non- f features. $\square$

B.2 Proof of Proposition C.1 (Dimensional Bound)

Proof. By Proposition 3.1, there exists a context-invariant subspace $\mathcal{S}_f$ containing all f -relevant information. We show that linear separability of K feature values constrains $\dim(\mathcal{S}_f) \leq K-1$.

Let $\{\mathbf{h}_{y_k}\}_{k=1}^K \subset \mathcal{S}_f$ be representative embeddings (e.g., class centroids) for each feature value y_k . For these to be linearly separable, they must be affinely independent: no $\mathbf{h}_{y_k}$ lies in the affine hull of the others.

The affine hull of K points has dimension at most $K-1$. To see this, define centered representations:

$$\tilde{\mathbf{h}}_{y_k} = \mathbf{h}_{y_k} - \bar{\mathbf{h}}, \quad \text{where } \bar{\mathbf{h}} = \frac{1}{K} \sum_{k=1}^K \mathbf{h}_{y_k} \quad (18)$$

These centered vectors satisfy $\sum_k \tilde{\mathbf{h}}_{y_k} = \mathbf{0}$, so at most $K-1$ are linearly independent. Thus $\text{span}(\{\tilde{\mathbf{h}}_{y_k}\}) \leq K-1$.

Since linear separability requires the f -relevant subspace to distinguish all K values, and this can be achieved in $K-1$ dimensions, the minimal invariant subspace satisfies $\dim(\mathcal{S}_f) \leq K-1$.

For binary features ($K=2$), this bound yields $\dim(\mathcal{S}_f) \leq 1$. Combined with the requirement that $\mathcal{S}_f$ be non-trivial (the feature must be detectable), we have $\dim(\mathcal{S}_f) = 1$, establishing directional invariance. $\square$

B.3 Proof of Proposition A.27 (Architectural Admissibility)

Proof. 2-layer case. For a 2-layer MLP $\text{MLP}(\mathbf{x}) = W_{\text{out}} \cdot \sigma(W_{\text{in}}\mathbf{x} + \mathbf{b}_{\text{in}}) + \mathbf{b}_{\text{out}}$, expand the matrix product directly:

$$\text{MLP}(\mathbf{x}) - \mathbf{b}_{\text{out}} = W_{\text{out}} \cdot \sigma(W_{\text{in}}\mathbf{x} + \mathbf{b}_{\text{in}}) = \sum_{i=1}^m \sigma(\langle \mathbf{w}_i^{\text{in}}, \mathbf{x} \rangle + b_{\text{in},i}) \cdot \mathbf{w}_i^{\text{out}}. \quad (19)$$

Setting $g_i(\mathbf{x}) := \sigma(\langle \mathbf{w}_i^{\text{in}}, \mathbf{x} \rangle + b_{\text{in},i})$, we have $\text{MLP}(\mathbf{x}) = \sum_i g_i(\mathbf{x}) \mathbf{w}_i^{\text{out}} + \mathbf{b}_{\text{out}}$, identifying $\mathbf{d}_i = \mathbf{w}_i^{\text{out}}$ and $m_i(\mathbf{x}) = g_i(\mathbf{x})$.

Verification of Definition A.25. (a) Context-independence: each write direction $\mathbf{w}_i^{\text{out}}$ is a column of W_{out} , a fixed model parameter, so $\mathbf{d}_i$ does not depend on $\mathbf{x}$. (b) Independent magnitudes: each gate $g_i(\mathbf{x})$ is computed by applying σ to a single linear projection $\langle \mathbf{w}_i^{\text{in}}, \mathbf{x} \rangle$. Because σ is applied per

coordinate, the value of $g_j(\mathbf{x})$ for $j \neq i$ does not enter the computation of $g_i(\mathbf{x})$, so the magnitudes are determined independently across features.

Deeper case ($L > 2$). For an L -layer MLP with weights $W_1, \dots, W_L$ and biases $\mathbf{b}_1, \dots, \mathbf{b}_L$:

$$\text{MLP}_L(\mathbf{x}) = W_L \cdot \sigma(W_{L-1} \cdot \sigma(\dots \sigma(W_1 \mathbf{x} + \mathbf{b}_1) \dots) + \mathbf{b}_{L-1}) + \mathbf{b}_L. \quad (20)$$

Write this as $\text{MLP}_L(\mathbf{x}) = W_L \cdot \mathbf{v}(\mathbf{x}) + \mathbf{b}_L$. Suppose, for contradiction, that this admits a selection-only decomposition with directions $\{\mathbf{d}_f\}$ taken from columns of W_L and scalars $\{m_f(\mathbf{x})\}$. Then condition (b) requires that the contribution along each column $\mathbf{d}_i$ depends on $\mathbf{x}$ only through a scalar $m_i(\mathbf{x})$ that is independent of the other gates' values.

But the i -th coordinate of $\mathbf{v}(\mathbf{x})$ is:

$$v_i(\mathbf{x}) = \sigma\left(\sum_j (W_{L-1})_{ij} \cdot u_j(\mathbf{x}) + (\mathbf{b}_{L-1})_i\right) \quad (21)$$

where each $u_j(\mathbf{x})$ is itself the post-nonlinearity activation of an upstream layer, and depends on multiple coordinates of yet-earlier layers' outputs. The scalar $v_i(\mathbf{x})$ is therefore a composite quantity that mixes the values of multiple intermediate gates: changing the value of $u_j(\mathbf{x})$ for $j \neq i$ generally changes $v_i(\mathbf{x})$ via the linear combination inside σ . Hence the magnitude associated with column i of W_L is coupled to the values of other intermediate activations and cannot be determined independently of other features' magnitudes. Condition (b) of Definition A.25 therefore fails. The same conclusion holds for any other choice of $\{\mathbf{d}_f\}$ inside the column span of W_L , since the coupling is intrinsic to the composition of multiple nonlinearities. $\square$

C Capacity Constraints and Feature Factorization

C.1 Detailed Argument for Capacity-Driven Feature Sharing

We provide a complete argument that capacity constraints in transformers favor factorized representations with shared, invariant feature directions.

Proposition (Capacity Constraint Implies Feature Sharing, restated). Let $\mathcal{M}$ be a transformer with vocabulary $|\mathcal{V}|$ and hidden dimension d , where $|\mathcal{V}| \gg d$. Under the following conditions:

1. **Linear readout:** Token logits are computed via $\text{logit}_t = \mathbf{w}_t^\top \mathbf{h}(c)$
2. **Sparse activation:** Each context c expresses a small subset of all possible features
3. **Shared features:** Multiple tokens share semantic attributes

Then the optimal representation factorizes tokens as:

$$\mathbf{w}_t = \sum_{f \in F_t} \alpha_{t,f} \mathbf{d}_f \quad (22)$$

where F_t is the feature set for token t , and $\{\mathbf{d}_f\}$ are shared feature directions with $|F| \ll |\mathcal{V}|$.

Proof. We proceed in four steps, establishing that factorization is optimal under capacity constraints when features are sparse.

Step 1: Dimensional constraint. The hidden dimension d bounds the number of orthogonal directions available. With typical values $|\mathcal{V}| \approx 50,000$ and $d \approx 4,000$, at most d tokens can have mutually orthogonal representations in W_U . The remaining $|\mathcal{V}| - d$ tokens must share directions with others, which creates potential interference in the readout.

Step 2: Interference cost depends on co-occurrence. Suppose tokens t_1 and t_2 share a direction $\mathbf{d}$, i.e., both $\mathbf{w}_{t_1}$ and $\mathbf{w}_{t_2}$ have non-zero projections onto $\mathbf{d}$. Interference occurs when both tokens are relevant predictions for a context c : the hidden state $\mathbf{h}(c)$ cannot independently modulate the logits for t_1 and t_2 along their shared direction. The expected interference cost is:

$$\mathcal{L}_{\text{interference}}(t_1, t_2) \propto P(t_1, t_2 \text{ both relevant in } c) \cdot |\mathbf{w}_{t_1}^\top \mathbf{d}| \cdot |\mathbf{w}_{t_2}^\top \mathbf{d}| \quad (23)$$

For sparse features—those with low co-occurrence probability across the corpus—this interference cost is small.

Step 3: Factorization minimizes total representation cost. Consider the total cost of a representation as the sum of dimensional cost (number of directions used) and interference cost. Compare two strategies:

- *Unique directions:* Each token t receives a dedicated direction $\mathbf{d}_t$. This eliminates interference but requires $|\mathcal{V}|$ dimensions—infeasible when $|\mathcal{V}| \gg d$.
- *Factorized features:* Tokens share $|F|$ feature directions, where each token t is represented as a sparse combination of features $f \in F_t$. This requires only $|F|$ dimensions plus interference costs proportional to feature co-occurrence.

When semantic features are sparse and $|F| \ll |\mathcal{V}|$, the factorized strategy achieves lower total cost. Intuitively, “mammal” and “European” rarely co-occur as the dominant semantic features of a context, so tokens sharing these directions (e.g., “cat” and “France”) experience minimal interference despite their shared structure.

Step 4: Each factor is communicable and invariant. Under factorization, the logit computation for token t decomposes as:

$$\text{logit}_t = \mathbf{w}_t^\top \mathbf{h}(c) = \sum_{f \in F_t} \alpha_{t,f} \mathbf{d}_f^\top \mathbf{h}(c) = \sum_{f \in F_t} \alpha_{t,f} \phi_f(c) \quad (24)$$

where $\phi_f(c) = \mathbf{d}_f^\top \mathbf{h}(c)$ is the activation of feature f in context c . For the model to correctly predict token probabilities, the hidden state $\mathbf{h}(c)$ must encode each relevant feature f such that $\phi_f(c)$ reflects the appropriate magnitude. This imposes three requirements on each feature direction $\mathbf{d}_f$:

1. **Linearly decodable:** The feature activation is extracted via a linear projection $\mathbf{d}_f^\top \mathbf{h}(c)$.
2. **Multi-context:** The same direction $\mathbf{d}_f$ is used across all tokens t with $f \in F_t$ and all contexts c where feature f is relevant.
3. **Context-invariant:** The direction $\mathbf{d}_f$ is fixed in W_U and does not depend on c ; only the magnitude $\phi_f(c)$ varies with context.

These three properties—linear decodability, multi-context usage, and context-invariance—are precisely the conditions of Theorem 3.3. Therefore, each semantic factor $\mathbf{d}_f$ occupies an invariant subspace in the representation geometry. $\square$

Connection to prior work. This result connects to the superposition hypothesis [Elhage et al., 2022], which demonstrates that neural networks exploit high-dimensional geometry to represent more features than dimensions when features are sparse. Our contribution is to show that this superposition, combined with the linear readout constraint of transformers, shapes the specific geometric form of feature representations: fixed directions with context-dependent magnitudes. While Achille and Soatto [2018] establish that information-minimal representations are necessarily invariant, we characterize the *form* this invariance takes within transformer architectures.

C.2 Proof of Proposition A.22

Proof. We analyze each component of Layer Normalization.

Step 1: Mean-centering. Define the mean-centering projection

$$\Pi_{\mathbf{1}^\perp} := I - \frac{1}{d} \mathbf{1} \mathbf{1}^\top,$$

where $\mathbf{1} \in \mathbb{R}^d$ is the all-ones vector. Applying this linear operator to our decomposition:

$$\Pi_{\mathbf{1}^\perp} h(c) = \alpha_f(c) \Pi_{\mathbf{1}^\perp} \mathbf{d}_f + \Pi_{\mathbf{1}^\perp} \eta(c).$$

Since $\mathbf{d}_f \notin \text{span}(\mathbf{1})$ by assumption, the projected direction $\mathbf{d}'_f := \Pi_{\mathbf{1}^\perp} \mathbf{d}_f \neq 0$.

Step 2: Scalar normalization. LayerNorm divides by the standard deviation $\sigma(h(c)) > 0$:

$$\frac{\Pi_{\perp} h(c)}{\sigma(h(c))} = \frac{\alpha_f(c)}{\sigma(h(c))} d'_f + \frac{\Pi_{\perp} \eta(c)}{\sigma(h(c))}.$$

Since $\sigma(h(c))$ is a positive scalar, this operation preserves the subspace $\text{span}(d'_f)$.

Step 3: Learned affine transformation. LayerNorm applies elementwise scaling by $\gamma \in \mathbb{R}^d$ and adds bias $\beta \in \mathbb{R}^d$:

$$\text{LN}(h(c)) = \frac{\alpha_f(c)}{\sigma(h(c))} (\gamma \odot d'_f) + \frac{\gamma \odot \Pi_{\perp} \eta(c)}{\sigma(h(c))} + \beta.$$

Define the transformed feature direction

$$\tilde{d}_f := \gamma \odot d'_f = \gamma \odot \Pi_{\perp} d_f.$$

This direction depends only on d_f and the learned parameter γ , and is therefore context-invariant.

LayerNorm maps the context-invariant subspace $\text{span}(d_f)$ to a new context-invariant subspace $\text{span}(\tilde{d}_f)$ via the fixed transformation $d_f \mapsto \gamma \odot \Pi_{\perp} d_f$. The feature remains affinely decodable: a linear probe along $\tilde{d}_f$ with appropriate bias recovers a signal proportional to $\alpha_f(c)$, scaled by $1/\sigma(h(c))$. $\square$

C.3 Proof of Proposition C.1 (Dimensional Bound)

Proposition C.1 (Dimensional Bound). *If f takes K linearly separable values, then $\dim(\mathcal{S}_f) \leq K - 1$. For binary features, this yields directional invariance: $\mathcal{S}_f = \text{span}(\{\mathbf{d}_f\})$.*

C.4 Proof of Corollary A.29 (Approximate Self-Reference)

Corollary (Approximate Self-Reference, restated). Let t_f be a token or expression that canonically expresses feature f . Assume its activation decomposes as $\mathbf{h}_{t_f} = \beta_f \mathbf{d}_f + \mathbf{u}_f$, where $\|\mathbf{d}_f\| = 1$, $\beta_f > 0$, and $\|\mathbf{u}_f\| \leq \delta_f < \beta_f$. Then the normalized class-prompt activation $\hat{\mathbf{h}}_{t_f} := \mathbf{h}_{t_f} / \|\mathbf{h}_{t_f}\|$ satisfies

$$\cos(\hat{\mathbf{h}}_{t_f}, \mathbf{d}_f) \geq \frac{\beta_f - \delta_f}{\beta_f + \delta_f}.$$

Proof. By Definition A.2, the activation $\mathbf{h}_{t_f}$ is the hidden state at some layer and position obtained by running the token (or expression) t_f through the model; it is an element of the context set $\mathcal{C}$. Because t_f canonically expresses f , applying Assumption A.12 to this context, the activation admits a sparse decomposition with $f \in S(t_f)$ and $\alpha_f(t_f) \neq 0$. Identifying $\beta_f := \alpha_f(t_f)$ (assumed positive without loss of generality, since the sign of $\mathbf{d}_f$ is a convention) and $\mathbf{u}_f := \sum_{g \in S(t_f), g \neq f} \alpha_g(t_f) \mathbf{d}_g + \boldsymbol{\xi}(t_f)$ as the residual contribution from non- f active features and noise, the hypothesis $\|\mathbf{u}_f\| \leq \delta_f$ corresponds to a bound on the magnitude of these other components in the class prompt's activation.

For the numerator,

$$\langle \mathbf{h}_{t_f}, \mathbf{d}_f \rangle = \beta_f + \langle \mathbf{u}_f, \mathbf{d}_f \rangle \geq \beta_f - \|\mathbf{u}_f\| \|\mathbf{d}_f\| \geq \beta_f - \delta_f, \quad (25)$$

using Cauchy–Schwarz and $\|\mathbf{d}_f\| = 1$.

For the denominator,

$$\|\mathbf{h}_{t_f}\| = \|\beta_f \mathbf{d}_f + \mathbf{u}_f\| \leq \beta_f + \|\mathbf{u}_f\| \leq \beta_f + \delta_f \quad (26)$$

by the triangle inequality. Combining,

$$\cos(\hat{\mathbf{h}}_{t_f}, \mathbf{d}_f) = \frac{\langle \mathbf{h}_{t_f}, \mathbf{d}_f \rangle}{\|\mathbf{h}_{t_f}\|} \geq \frac{\beta_f - \delta_f}{\beta_f + \delta_f}. \quad (27)$$

The bound is positive whenever $\delta_f < \beta_f$, which is the condition that the target-feature component dominates the residual. Note that this proof does not single out any specific interface or layer: it applies wherever Assumption A.12 applies, so class-prompt activations can be extracted from any layer or attention head and used as approximate references; the choice of extraction point is an empirical matter. $\square$

C.5 Additional Theoretical Implications

Our framework yields several additional consequences for transformer interpretability:

Corollary C.2 (Attention Preserves Feature Identity). *The OV circuit linearly transforms features while preserving their identity. If position j encodes feature f as $\mathbf{h}_j = \alpha_f \mathbf{d}_f + \boldsymbol{\eta}_j$, the attention output at position i is:*

$$\mathbf{o}_i^{(f)} = \underbrace{\left(\sum_j a_{ij} \cdot \alpha_f^{(j)} \right)}_{\text{context-dependent magnitude}} \cdot \underbrace{W_O W_V \mathbf{d}_f}_{\text{linearly transformed direction}} \quad (28)$$

The QK circuit modulates how much of the feature transfers (magnitude); the OV circuit determines how the feature direction is transformed.

Proof. By linearity of the OV circuit:

$$\mathbf{o}_i = \sum_j a_{ij} \cdot W_O W_V \mathbf{h}_j \quad (29)$$

$$= \sum_j a_{ij} \cdot W_O W_V (\alpha_f^{(j)} \mathbf{d}_f + \boldsymbol{\eta}_j) \quad (30)$$

$$= \left(\sum_j a_{ij} \cdot \alpha_f^{(j)} \right) W_O W_V \mathbf{d}_f + \sum_j a_{ij} \cdot W_O W_V \boldsymbol{\eta}_j \quad (31)$$

The f -relevant component is the first term, with magnitude determined by attention-weighted sum and direction determined by linear transformation of $\mathbf{d}_f$. $\square$

Corollary C.3 (Distributional Influence). *Adding a feature direction $\mathbf{d}_f$ to the hidden state shifts the output distribution predictably:*

$$\Delta \text{logit}_t = \lambda \cdot \mathbf{w}_t^\top \mathbf{d}_f \quad (32)$$

Tokens aligned with $\mathbf{d}_f$ are boosted; anti-aligned tokens are suppressed. This provides the theoretical foundation for activation steering [Turner et al., 2023, Zou et al., 2023].

Proof. Let $\mathbf{h}$ be the original hidden state and $\mathbf{h}' = \mathbf{h} + \lambda \mathbf{d}_f$ the steered state. By linearity of unembedding:

$$\text{logit}_t(\mathbf{h}') = \mathbf{w}_t^\top \mathbf{h}' = \mathbf{w}_t^\top \mathbf{h} + \lambda \mathbf{w}_t^\top \mathbf{d}_f = \text{logit}_t(\mathbf{h}) + \lambda \mathbf{w}_t^\top \mathbf{d}_f$$

Thus $\Delta \text{logit}_t = \lambda \mathbf{w}_t^\top \mathbf{d}_f$, which is positive when $\mathbf{w}_t$ and $\mathbf{d}_f$ are aligned. $\square$

Corollary C.4 (Non-Bidirectionality). *The invariant subspace for “A predicts B” is generally distinct from that for “B predicts A”:*

$$\mathcal{S}_{A \rightarrow B} \not\subseteq \mathcal{S}_{B \rightarrow A} \quad \text{and} \quad \mathcal{S}_{B \rightarrow A} \not\subseteq \mathcal{S}_{A \rightarrow B} \quad (33)$$

This provides a geometric explanation for the reversal curse [Berglund et al., 2023].

Proof. The features “A predicts B” and “B predicts A” are distinct communicable features—they produce different outputs and are learned from different training examples. By Proposition 3.1, each occupies its own invariant subspace determined by that feature. Since these are different features, their subspaces need not coincide. Models trained on “A is B” learn $\mathcal{S}_{A \rightarrow B}$ but have no reason to simultaneously learn $\mathcal{S}_{B \rightarrow A}$, explaining the failure to infer reverse relations. $\square$

D Additional Testing on Unsupervised Probes

D.1 Unsupervised Probes

We validate these properties through comprehensive visualization analysis, comparing embeddings before and after probe transformation using both PCA and t-SNE projections. These visualizations confirm that unsupervised probes learn transformations that improve class separability—instances cluster more tightly around their corresponding prototypes in the transformed space—while the prototype vectors themselves maintain interpretable directional structure aligned with semantic features.

Zero-shot generalization to unseen classes. PCA (top) and t-SNE (bottom) projections of month embeddings after unsupervised probe transformation. Training used only odd months; even months were held out. In both visualizations, held-out classes cluster tightly around their respective prototypes, demonstrating that the learned geometric structure generalizes without exposure to these classes during training.

E Additional Experiments

This section reports additional experiments that validate the robustness, statistical significance, and scope of the framework on more complex tasks and across different computational pathways.

E.1 Relation between class vectors and instance vectors

E.2 Polysemy as the inverse of *Partial Synonyms*

Polysemy provides a natural test case for the directional account: “Apple” encodes both fruit and company meanings as superposed directions. Table 1 shows both Fruits and Companies datasets classify accurately despite sharing this token. To test explicit disentanglement, we combine both datasets and compare using a single “Apple” token versus distinct tokens (“Fruit apple”, “Company Apple”). Both strategies achieve comparable accuracy (Fig. 8: 69% vs. 65.7%), confirming that both meanings co-exist in superposition—context modulates magnitude, not selection.

This geometric view inverts naturally: “Dog” and “Cat” are distinct tokens sharing a component along $\mathbf{d}_{\text{mammal}}$ —*partial synonyms* with respect to that feature. Polysemy (one token, multiple directions) and partial synonymy (multiple tokens, shared direction) are two sides of the same coin.

E.3 Cross-Architecture Validation

We evaluate three architectures outside the decoder-only LLM regime: BERT (encoder-only, masked language modeling), ViT (vision-only encoder), and LLaVA (vision-language). For BERT we apply the zero-shot probe to a subset of the text tasks. For ViT, which has no text vocabulary, we use the [CLS] representation of a canonical image as the class direction. For LLaVA we use object names as class directions and images as instances. ViT and LLaVA are evaluated on a 10-class, 1000-image ImageNet subset.

Table 2: Cross-architecture validation. The Self-Reference Property holds across encoder-only (BERT), vision-only (ViT), and vision-language (LLaVA) transformers, supporting the prediction that linear interfaces—rather than autoregressive language modeling—are what drives invariant subspace structure.

Model	Modality	Task	Acc.	Random
BERT	text (encoder)	Languages	93.0%	16.7%
BERT	text (encoder)	Countries	61.5%	20.0%
BERT	text (encoder)	Emotions	48.9%	16.7%
ViT	vision	ImageNet-10	79.5%	10.0%
LLaVA	vision-language	ImageNet-10	78.6%	10.0%

All five settings recover the invariant structure well above the random baseline. BERT—trained with masked language modeling rather than next-token prediction—reaches 93% on Languages, comparable to the decoder-only models in Table 1. The vision and vision-language results are more

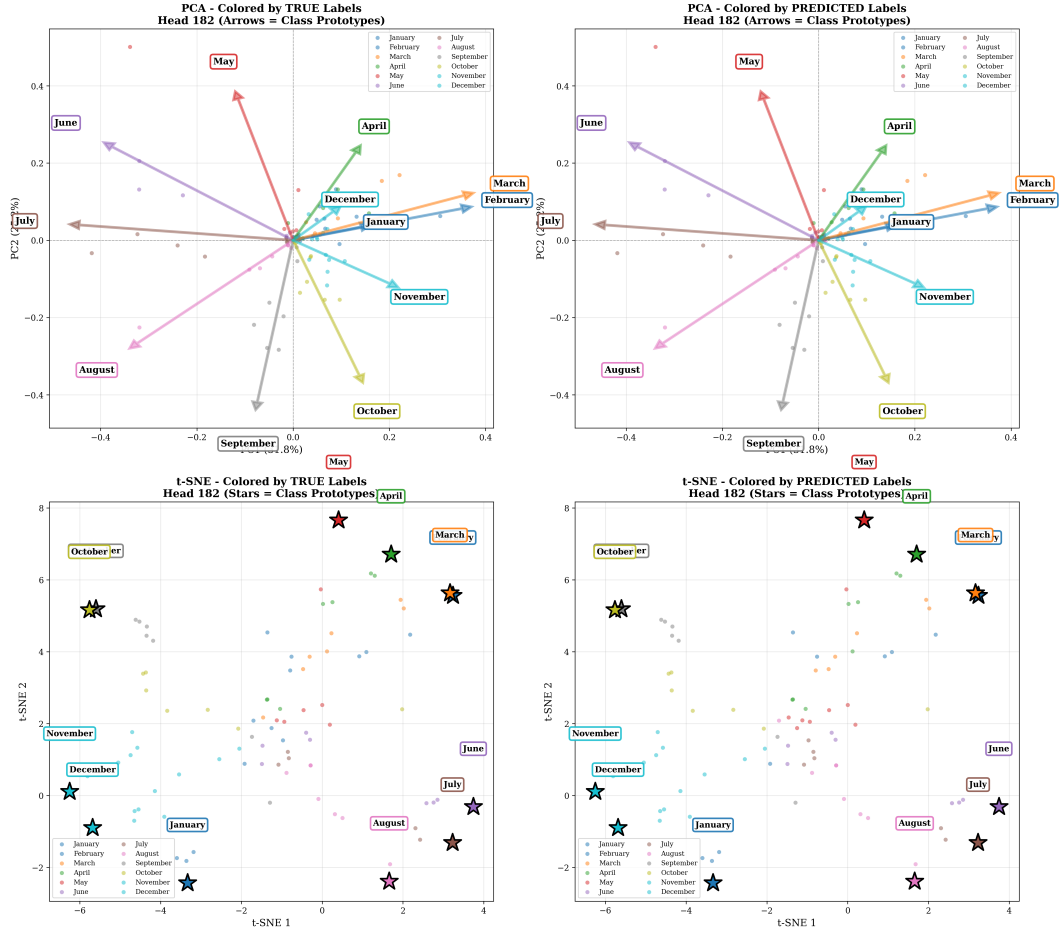


Figure 5: Zero-shot generalization of unsupervised probes. Probes trained only on odd months (Jan, Mar, May, Jul, Sep, Nov) successfully classify held-out even months, demonstrating that the learned transformation captures generalizable temporal structure rather than memorizing training classes.

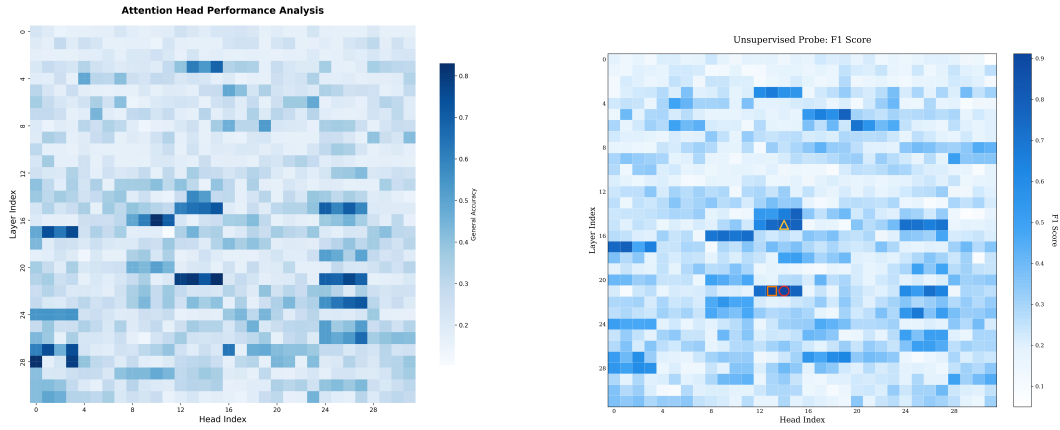


Figure 6: Zero-shot (left) and Unsupervised prob classification performance on the Animal dataset in Llama-3-8B. Both contain similar accuracy distributions.

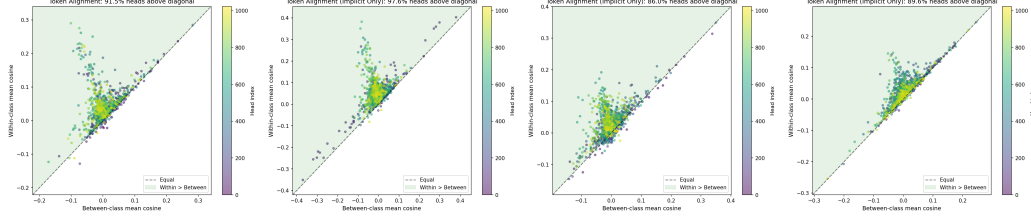


Figure 7: Token alignment validation of the Self-Reference Property across four datasets in LLaMA3-8B. Each point represents one attention head; the x-axis shows mean cosine similarity between class tokens and *other*-class implicit instances, while the y-axis shows similarity to *same*-class implicit instances. Points above the diagonal indicate stronger alignment with the correct class. Percentages indicate heads above diagonal: **Countries** 91.5%, **Animals** 97.6%, **Cartoon Characters** 86.0%, **Emotions** 89.6%. Heads below the diagonal are consistent with head specialization: not every head is expected to encode the class-relevant feature, and some heads may encode shared or opposing subfeatures.

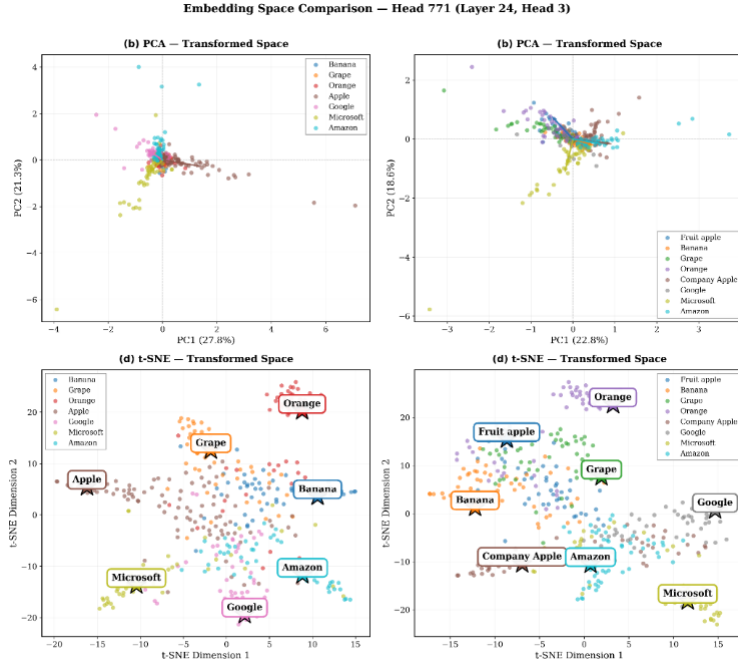


Figure 8: PCA and t-SNE projections of embeddings for the polysemous word Apple (fruit vs. company) using domain-specific tokens. (Left) A single “Apple” token representing both classes (69% accuracy). (Right) Separate tokens “Fruit apple” and “Company Apple” treated as distinct classes (65.7% accuracy). Both methods successfully disentangle the two senses, with instances clustering around their respective class prototypes.

striking: ViT has no text interface at all, yet the [CLS] image representation provides a class direction that linearly separates ImageNet classes at roughly $8\times$ random.

E.4 Compositional and Multi-hop Task Details

This appendix gives the full task specifications and per-task results for the compositional and multi-hop experiments summarized in Section 5.

Multi-hop chain. Novel words are mapped to categories in-context, and items pass through ownership chains. The model must resolve: person $\rightarrow$ person $\rightarrow$ novel word $\rightarrow$ category. Example ($k=2$): “A blorp is a fruit. [...] Dave bought a blorp. Dave gave his item to Eve. What food group

Table 3: Compositional and multi-hop tasks. “Best Head” is the accuracy of the best attention head; “Ratio” is best-head accuracy divided by the random baseline. Individual operation F1 reaches 0.89–1.0 in the ICL operator tasks.

Task	Cls.	Gen.	B. Head	Rand.	Ratio
Multi-hop chain ($k=1$)	8	79%	74%	12.5%	5.9×
Multi-hop chain ($k=2$)	8	45%	39%	12.5%	3.1×
Multi-hop chain ($k=3$)	8	35%	35%	12.5%	2.8×
Language translation	5	31%	58%	20%	2.9×
Vowel append	5	71%	61%	20%	3.1×
Text operations	5	82%	48%	20%	2.4×
Arithmetic (1-step add)	27	88%	86%	3.7%	23×
Arithmetic (3-step add)	27	58%	66%	3.7%	18×
Arithmetic (1-step mul)	27	92%	64%	3.7%	17×
Arithmetic (2-step mul)	27	78%	56%	3.7%	15×
Arithmetic (mul & add)	27	6%	24%	3.7%	6.5×
SVAMP word problems	60	53%	45%	1.7%	26×
refCOCO	10	—	50%	10%	5×

does Eve have?” $\rightarrow$ *blorp*. Prototypes are the novel words (blorp, snark, ...), and mappings change every sample, preventing memorization.

ICL operator tasks. Arbitrary symbols are mapped to operations through few-shot examples. We extract activations at each operator-token position and classify against prototypes (5-way). Three variants:

- *Language translation* (*=Spanish, #=French, ~ =Portuguese, +=German, @=Italian): “apple #tree $\rightarrow$ manzana arbre”. Prototypes: language names. The probe recovers language identity (individual F1 up to 1.0) even when the model only generates correct translations 31% of the time.
- *Vowel append* (*=a, #=e, ~ =i, +=o, @=u): “dog #cat $\rightarrow$ doga cate”. Prototypes: single-vowel tokens.
- *Text operations* (*=capitalize, #=plural, +=repeat, ~ =suffix, @=truncate): “*apple #tree $\rightarrow$ APPLE trees”. Evaluated at the operator-token positions.

Arithmetic composition. Does the token “13” (without context) provide the direction that aligns with the model’s representation when computing “4+9=”? Prototypes are number tokens 3–30. We tested addition and multiplication, restricting to whole-number answers.

SVAMP word problems. Natural-language math requiring multi-step reasoning. We use numbers ≤ 60 as answer classes.

refCOCO. Object description and image. Using LLaVA, we pair each image with its correct referring expression and 9 descriptions from random images (10 classes, random = 10%). We perform inference with a placeholder prompt and each text description separately, extract attention-head outputs at the last-token position, and classify by projection similarity.

E.5 Robustness to Input Perturbations

We evaluate stability under three types of perturbations: (1) syntax-level shuffling, (2) random-noise injection, and (3) adversarial system-level prefixes. As shown in Table 4, all methods behave similarly under syntax and noise perturbations, indicating robustness to surface-level variations. Under adversarial prefixes, however, a difference emerges: on Countries, LLM accuracy drops sharply (0.79 $\rightarrow$ 0.39), while the Unsupervised Probe remains stable (0.80 $\rightarrow$ 0.81). This suggests that our method is less sensitive to prompt manipulation, which we attribute to its reliance on representation-level geometry rather than autoregressive decoding.

Table 4: Robustness under input perturbations. The Unsupervised Probe is more stable than autoregressive decoding under adversarial system prefixes.

Method	Base	Syntax	Noise	System
LLM output	0.74	0.66	0.69	0.71
Zero-shot probe	0.71	0.66	0.68	0.70
Unsupervised	0.76	0.67	0.68	0.76

E.6 Statistical Significance over Multiple Seeds

We repeat the classification experiments of Table 1 with 20 independent random seeds and report the mean, standard deviation, and 95% confidence intervals. As shown in Table 5, variance is consistently low and confidence intervals are tight (± 0.002 – 0.018) across all tasks and model families, indicating stable performance across runs. The observed improvements of the Unsupervised Probe are not due to stochastic variation but reflect a consistent property of the learned representations.

Table 5: Mean accuracy with standard deviation and 95% confidence intervals over 20 seeds. Each cell shows ACC / Std / CL.

Task	LLaMA-8B	LLaMA-3B	Mistral	GPT2-small
Animals	0.94 / 0.004 / ± 0.002	0.89 / 0.010 / ± 0.005	0.87 / 0.012 / ± 0.006	0.29 / 0.016 / ± 0.008
Countries	0.81 / 0.018 / ± 0.009	0.82 / 0.038 / ± 0.018	0.84 / 0.025 / ± 0.012	0.52 / 0.037 / ± 0.018
Characters	0.60 / 0.013 / ± 0.006	0.47 / 0.016 / ± 0.008	0.54 / 0.010 / ± 0.005	0.26 / 0.017 / ± 0.008
Authors	0.76 / 0.010 / ± 0.005	0.59 / 0.015 / ± 0.007	0.53 / 0.015 / ± 0.007	0.21 / 0.014 / ± 0.007
Languages	0.89 / 0.012 / ± 0.006	0.92 / 0.011 / ± 0.006	0.98 / 0.004 / ± 0.002	0.83 / 0.008 / ± 0.004
Emotions	0.54 / 0.007 / ± 0.003	0.53 / 0.006 / ± 0.003	0.51 / 0.007 / ± 0.003	0.22 / 0.010 / ± 0.005
Fruits	0.70 / 0.014 / ± 0.007	0.65 / 0.017 / ± 0.008	0.64 / 0.012 / ± 0.006	0.39 / 0.017 / ± 0.008
Companies	0.82 / 0.011 / ± 0.005	0.77 / 0.009 / ± 0.004	0.83 / 0.005 / ± 0.003	0.61 / 0.013 / ± 0.006

E.7 Attention vs. MLP Outputs

To probe the boundary of the linear-interface assumption, we compare attention outputs (linear) against MLP activations (non-linear) on the same classification tasks. Attention consistently outperforms MLP on categorical concepts (Animals, Countries, Emotions). For more complex concepts (Authors, Characters), MLP achieves comparable performance but with less consistency across tasks. We probe MLP outputs only at the last-token position, which may underestimate their performance. These results suggest that MLP outputs, while computed nonlinearly, still produce linearly accessible features in the residual stream—consistent with the fact that MLP outputs are added linearly to the residual stream and must be readable by downstream linear interfaces. The corresponding numbers are reported as the bottom block of Table 1 in the main text. A more systematic analysis of which feature types admit such behavior (e.g., compositional or abstract reasoning features) is an important direction for future work.

F Supplementary Visualizations

This section provides additional SAE-based analyses supporting the invariant feature sharing and head-level sparsity patterns discussed in the main paper.

Across different datasets and models, we report (i) shared top- k SAE latent dimensions between instance-mean and class-prototype representations within individual heads, and (ii) unsupervised head-level classification performance using SAE latents. Figures are grouped by dataset and model to demonstrate the consistency of these patterns.

F.1 Dataset Explanation

We constructed several datasets to evaluate where semantic attributes are encoded in LLMs and whether semantic features exhibit this linear invariance.

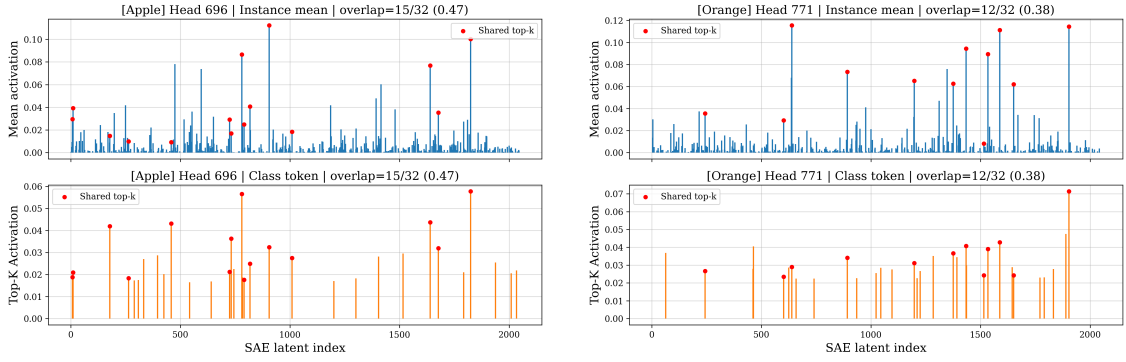


Figure 9: Shared top- k SAE latent activations between instance-mean and class-prototype representations for the Apple and Orange classes in Llama-3-8B, highlighting class-consistent sparse features within individual attention heads.

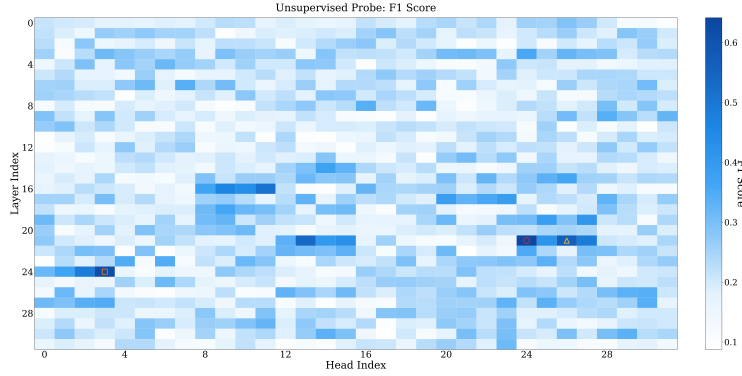


Figure 10: Unsupervised head-level classification performance on the Fruits task using SAE latent representations extracted from Llama-3-8B.

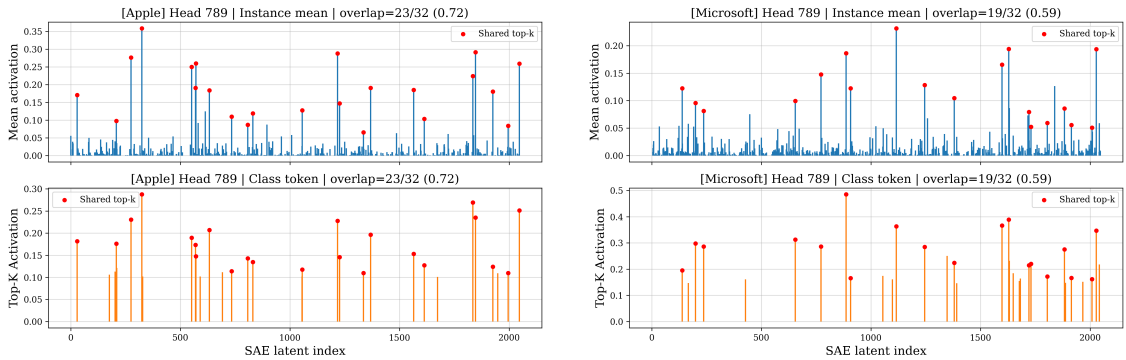


Figure 11: Shared top- k SAE latent activations between instance-mean and class-prototype representations for the Apple and Microsoft classes in Mistral, indicating head-specific sparse features aligned with company semantics.

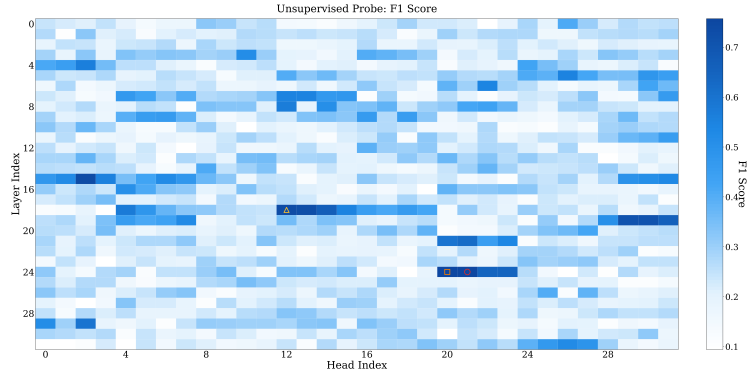


Figure 12: Unsupervised head-level classification performance on the Companies task using SAE latent representations extracted from Mistral.

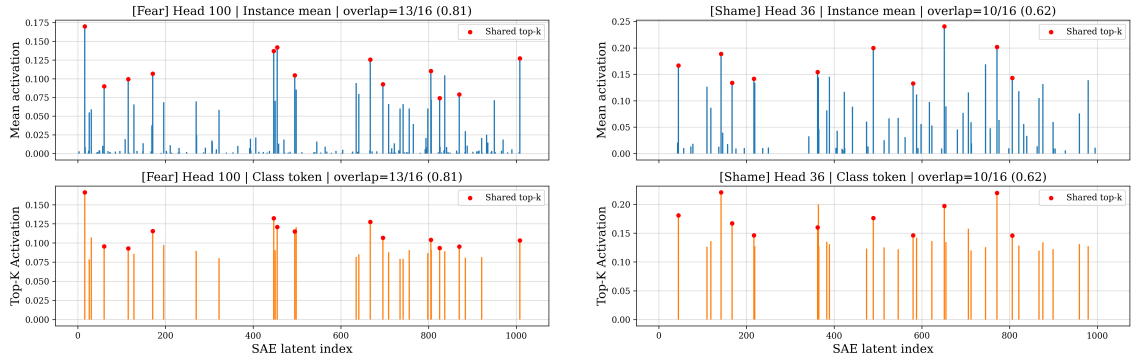


Figure 13: Shared top- k SAE latent activations between instance-mean and class-prototype representations for the Fear and Shame emotion classes in GPT-2.

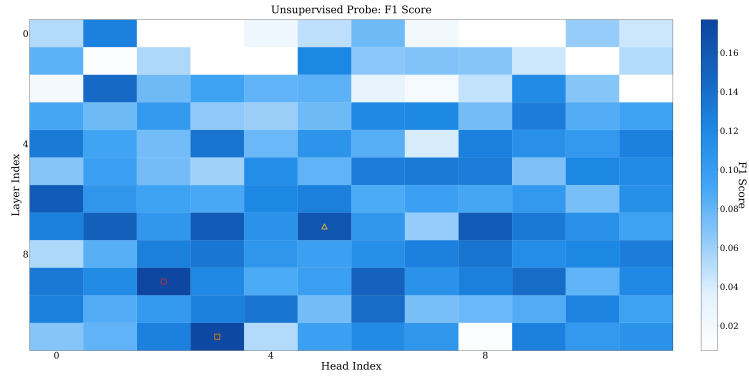


Figure 14: Unsupervised head-level classification performance on the Emotions task using SAE latent representations extracted from GPT-2.

Dataset	Classes	Instances/Class
Animals	6	50
Countries	5	39
Emotional Sentences	6	60
Literary Quotes	6	50
Cartoon Phrases	6	50
Languages	6	50
Fruits	4	50
Companies	4	50

Table 6: Summary of task-specific prompt datasets used for probing invariant feature directions across semantic, stylistic, linguistic, and affective domains.

Countries. We analyze the spaces of all the countries in the world grouped by their respective continent.

Languages. To evaluate cross-lingual representations, we employed the XQuAD dataset [Artetxe et al., 2020], a benchmark for cross-lingual question answering. We sampled 200 questions per language, covering English, Spanish, Russian, Hindi, German, and Mandarin Chinese.

Emotions. We used a subset of the Emotion Cause dataset [Ghazi et al., 2015], which includes 1,594 English sentences annotated with seven emotion labels (fear, sadness, anger, happiness, surprise, disgust, and shame). We sampled 600 examples for evaluation.

Cartoon Characters. To probe stylistic and character-specific features, we constructed a dataset centered on fictional characters with distinctive linguistic patterns. For each of six characters—Elmer Fudd, Foghorn Leghorn, Jar Jar Binks, Porky Pig, Scooby-Doo, and Yoda—we collected 50 iconic phrases from publicly available sources.

Literary Authors. We obtained 20 book quotes per author—William Faulkner, Gabriel García Márquez, Ernest Hemingway, Edgar Allan Poe, Virginia Woolf, William Shakespeare, and Mark Twain—from Goodreads.

Animals. This dataset includes animal species names categorized by biological class: mammals, invertebrates, birds, amphibians, reptiles, and fish.

Fruits and Companies. To test polysemy, we compiled a descriptive dataset where tokens like “Apple” belong to both the fruit and company categories, enabling evaluation of how context disambiguates shared representational directions. The company classes were “Apple”, “Google”, “Microsoft”, and “Amazon”, while the fruit classes were “Apple”, “Grape”, “Orange”, and “Banana”..

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Justification: Section 4 describes the contrastive training objective for the Unsupervised Probe and SAE training following Kissane et al. [2024]; Section 5.1 specifies the grokking-experiment setup ($p = 97$ modular division, 2-layer transformer, MLP classification head); Appendix F.1 reports per-task dataset sizes and class structure.

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Justification: Appendix E.6 reports mean accuracy, standard deviation, and 95% confidence intervals over 20 independent random seeds for all classification tasks across four model families.

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